國立臺灣科技大學

工業管理系

碩士學位論文

學號：M11301201

預測不確定性下基於分布穩健強化學習之微電網電池儲能系統排程研究

Distributionally Robust Reinforcement Learning for Battery Energy Storage System Scheduling in Microgrids under Forecast Uncertainty

研　究　生：陳柏穎

指導教授：周碩彥博士

　　　　　郭伯勳博士

中華民國 115 年 6 月

指導教授推薦書

學位考試委員審定書

預測不確定性下基於分布穩健強化學習之微電網電池儲能系統排程研究

研　究　生　：　陳　柏　穎

指　導　教　授　：　周　碩　彥

時　間　：　115　年　6　月

論文摘要

本研究針對國立臺灣科技大學校園微電網，提出基於分布穩健強化學習之能源管理策略，依據負載需求、太陽能發電量、電池狀態與時間電價動態調度電池儲能系統，以降低總營運成本並抑制尖峰購電需求。

針對混合整數線性規劃排程依賴預測資訊且易受預測誤差影響之限制，本研究採用以 Soft Actor-Critic (SAC)為基礎所發展的 Distributionally Robust Soft Actor-Critic（DR-SAC）作為核心方法，透過與環境互動學習充放電策略，並納入狀態轉移分布不確定性以提升排程穩健性。

為驗證效能，本研究以混合整數線性規劃之理論最佳解為基準，比較 SAC 與 DR-SAC 之排程結果，並導入具預測誤差之排程方法分析不確定性對成本之影響。系統模型納入時間電價、契約容量成本、台灣再生能源憑證收益與電池退化成本。結果顯示，DR-SAC 相較於 SAC 可降低總營運成本、契約容量相關成本與電池退化成本，並在預測不確定性下維持穩定且具競爭力之排程表現。

關鍵字：分布穩健強化學習、電池儲能系統、微電網、能源管理、預測不確定性

Distributionally Robust Reinforcement Learning for Battery Energy Storage System Scheduling in Microgrids under Forecast Uncertainty

Abstract

This study proposes an energy management strategy based on distributionally robust reinforcement learning for the campus microgrid of National Taiwan University of Science and Technology. The strategy dynamically dispatches a battery energy storage system according to load demand, photovoltaic generation, battery state of charge, and time-of-use pricing, to reduce total operating cost and suppress peak power demand.

To address the limitations of mixed-integer linear programming (MILP) scheduling, which relies on forecast information and is vulnerable to prediction errors, this study adopts Distributionally Robust Soft Actor-Critic (DR-SAC), an extension of Soft Actor-Critic (SAC), as the core method. The agent learns charge and discharge policies through environment interaction and incorporates state transition distribution uncertainty to enhance scheduling robustness.

For performance evaluation, a MILP model is formulated as a theoretical optimum under perfect information. SAC and DR-SAC are compared to assess the distributionally robust architecture, and a forecast-dependent method with prediction errors is introduced to analyze uncertainty's effect on cost. The system model incorporates time-of-use pricing, contract capacity charges, Taiwan Renewable Energy Certificate revenue, and battery degradation cost. Results show that DR-SAC reduces total operating cost, contract capacity-related cost, and battery degradation cost compared to SAC, while maintaining stable and competitive scheduling performance under forecast uncertainty.

Keywords: Distributionally Robust Reinforcement Learning, BESS, Microgrid, Energy Management, Forecast Uncertainty

Acknowledgement

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1
INTRODUCTION

1.1 Background

As the integration of renewable energy and building energy management requirements continue to grow, microgrids have emerged as an important architecture for distributed energy systems [1]. For microgrids equipped with photovoltaic (PV) generation and battery energy storage systems (BESS), the core challenge of energy management lies not only in utilizing renewable energy, but also in determining when to charge and discharge the battery across different time periods, so that load demand can be met while minimizing overall operating costs.

BESS plays a key role in regulating the mismatch between energy supply and demand in PV-BESS microgrids. When PV generation is high or electricity prices are low, BESS can store energy; when load demand is high or electricity prices are high, BESS can release energy to reduce grid purchases. Accordingly, BESS scheduling can be applied to energy time-shifting, peak demand control, and electricity cost reduction. However, if the scheduling strategy does not account for battery degradation, excessive charging and discharging may occur, leading to higher long-term degradation costs despite short-term cost savings. As a result, recent microgrid scheduling research has increasingly incorporated battery degradation cost into scheduling and cost evaluation frameworks to reflect the true economic value of BESS more accurately [2][3].

In practice, both PV generation and building load demand are subject to uncertainty. Conventional optimization-based scheduling methods typically rely on forecasted information such as future PV generation, load demand, or electricity prices [4]. When such information is derived from prediction models, scheduling performance may be affected by forecast errors. Even when a model yields an optimal schedule under predicted conditions, deviations between actual and forecasted operating conditions can result in additional electricity purchase costs, reduced peak control effectiveness, or lower battery utilization efficiency. Therefore, maintaining scheduling performance and stability under uncertain operating conditions remains an important challenge in PV-BESS microgrid energy management.

Reinforcement learning (RL) offers a sequential decision-making framework well-suited to the BESS scheduling problem. Unlike conventional scheduling methods that require complete future information in advance, an RL agent selects control actions based on the current system state and learns a long-term cost-minimizing policy through interaction with the environment. Recent studies have applied RL to real-time microgrid energy management and battery scheduling, demonstrating its ability to handle dynamic decision-making under renewable energy and load variability [5]. Among RL algorithms, Soft Actor-Critic (SAC) is suitable for BESS charge and discharge decisions due to its ability to handle continuous action spaces.

However, standard RL methods may still be sensitive to transition uncertainty in microgrid environments, where the same state-action pair can lead to different outcomes under varying operating conditions. This vulnerability may affect the stability of the learned policy when facing operating conditions that deviate from the training distribution. Recent research on distributionally robust optimization and distributionally robust RL has sought to address this issue by accounting for possible distribution shifts, thereby seeking a balance between cost performance and policy robustness [6][7].

Based on the above background, this study investigates BESS scheduling for a grid-connected PV-BESS microgrid, and examines the feasibility of incorporating distributionally robust concepts into the SAC framework. Using campus data from National Taiwan University of Science and Technology (NTUST) as a case study, this study establishes a total operating cost framework encompassing electricity purchase cost, contract capacity-related cost, battery degradation cost, and Taiwan Renewable Energy Certificate (TREC) revenue. SAC is adopted as the baseline method, and Distributionally Robust Soft Actor-Critic (DR-SAC) is applied for BESS scheduling, enabling the agent to make continuous charge and discharge decisions based on the current system state [8]. In addition, a Mixed Integer Linear Programming (MILP)-based theoretical optimum is constructed as an ideal benchmark under perfect future information, providing a reference for evaluating the cost gap between RL-based scheduling results and the optimal solution.

1.2 Objective

The primary objective of this study is to establish a BESS scheduling analysis framework for PV-BESS microgrids and to evaluate the impact of DR-SAC on scheduling cost and policy robustness under uncertain operating conditions. To this end, this study first establishes a PV-BESS microgrid scheduling framework that incorporates energy purchase cost, contract capacity cost, battery degradation cost, and TREC revenue into a consistent total operating cost structure. A MILP-based theoretical optimum model is then constructed to derive the optimal scheduling solution under complete future information, serving as a benchmark for evaluating the performance of all scheduling methods. Building on this foundation, SAC and DR-SAC are applied to the BESS scheduling problem, enabling the agent to make continuous charge and discharge decisions based on the current system state without relying on complete future information. Finally, the total operating cost, cost components, and BESS operation behavior of SAC, DR-SAC, and the theoretical optimum are compared to assess whether DR-SAC can improve cost performance and scheduling robustness relative to standard SAC.

1.3 Scope and Limitation

This study focuses on the BESS scheduling problem for a grid-connected PV-BESS microgrid, with a scheduling environment built on an hourly time step. The core of the research is to determine the charging, discharging, or idle operation of BESS at each time step to minimize the total operating cost over the testing period. Cost evaluation includes energy purchase cost, contract capacity cost, battery degradation cost, and TREC revenue [9], where battery degradation cost accounts for both calendar degradation and cycle degradation to reflect battery lifetime loss due to both the passage of time and actual charge and discharge operations [10]. This study compares three methods: MILP-based theoretical optimum, SAC, and DR-SAC. The theoretical optimum assumes that all exogenous data over the testing period are known in advance, and therefore serves only as an idealized benchmark rather than a directly deployable real-time scheduling method [11][12]. SAC serves as the standard RL baseline for learning a continuous BESS charge and discharge policy, while DR-SAC incorporates transition distribution uncertainty into the SAC framework to investigate the effect of distributionally robust formulation on scheduling robustness.

Regarding the limitations of this study, the simulation environment is constructed using historical data without actual field deployment, and therefore issues related to physical device control, communication latency, and hardware integration are not addressed. This study focuses on BESS scheduling rather than BESS sizing optimization, and battery capacity, rated power, and system equipment parameters are treated as given conditions. In addition, no PV generation or load demand forecasting model is developed, as the research emphasis is on scheduling methods under uncertainty rather than on forecasting models themselves. The MILP-based theoretical optimum relies on complete future information, and its results are used primarily to quantify the cost gap between RL-based scheduling methods and the theoretical optimal solution rather than to serve as a practical scheduling approach. Finally, this study focuses on cost performance and BESS operation behavior, and does not further address power quality, frequency control, voltage stability, multi-BESS coordination, or distribution network power flow.

1.4 Organization of Thesis

This thesis is organized into five chapters, with the content of each chapter described as follows.

Chapter 1 provides an introduction, presenting the research background, objectives, scope and limitations, and the overall structure of the thesis.

Chapter 2 presents a literature review, covering related research on microgrid energy management, forecast uncertainty, RL-based scheduling, robust energy scheduling, and battery degradation, and identifies the research gap addressed by this study.

Chapter 3 describes the methodology, explaining the PV-BESS microgrid system framework, mathematical formulation, MILP-based theoretical optimum, and the construction of SAC and DR-SAC for the BESS scheduling problem.

Chapter 4 presents the case study and result analysis, describing the NTUST campus dataset, experimental settings, and test results, and comparing SAC, DR-SAC, and the theoretical optimum in terms of total operating cost, cost components, and BESS operation behavior.

Chapter 5 presents the conclusions and directions for future research, summarizing the main findings of the study and discussing its limitations and potential extensions.

CHAPTER 2
LITERATURE REVIEW

2.1 Microgrid Energy Management

With the continuous increase in renewable energy penetration, microgrid systems have attracted growing attention as flexible and decentralized distributed energy management frameworks. A microgrid typically consists of distributed energy resources, energy storage systems, electricity loads, control systems, and a connection to the utility grid, enabling the local coordination of energy generation, storage, and consumption. Compared with conventional centralized power systems, microgrids can enhance renewable energy integration, improve power supply reliability, and provide more flexible energy dispatch [13][14]. Therefore, how to coordinate the operation of different energy components through an effective energy management system has become an important issue in microgrid research.

In microgrid energy management, the BESS is commonly regarded as an important flexibility resource. Due to the intermittency of PV generation and the variability of building load demand across time periods and usage patterns, there is often a temporal mismatch between energy supply and demand. BESS can store electricity during periods of low demand, low electricity prices, or high PV generation, and discharge during periods of high demand or high electricity prices. Through this operation, BESS can support energy time-shifting, electricity cost reduction, peak shaving, and improvement of renewable energy self-consumption [14][15]. Therefore, BESS is not only an energy storage device, but also a core dispatchable resource for achieving economic efficiency and operational flexibility in microgrid energy management.

Campus microgrids are one of the typical application scenarios in microgrid energy management research. Compared with residential users or single commercial buildings, campus environments usually consist of multiple buildings, diverse electricity consumption activities, and more evident daily and seasonal load variations. Previous studies have used university campuses as case study sites for microgrid planning, energy management, and BESS scheduling strategy validation, and have indicated that campus microgrids are valuable for evaluating renewable energy integration, electricity cost control, and peak demand management [16][17][18]. Therefore, campus-based PV-BESS microgrids can serve as suitable cases for analyzing the cost effectiveness and operational behavior of BESS scheduling methods under real operating conditions.

The economic benefits of BESS scheduling are further influenced by electricity tariff structures and demand-related charges. Under a Time-of-Use (TOU) pricing mechanism, electricity prices vary across different time periods, allowing BESS to reduce electricity purchase costs by charging during low-price periods and discharging during high-price periods. The three-period TOU tariff adopted in this study, based on the tariff structure of Taiwan Power Company [19], is shown in Table 2.1. The tariff rates differ between summer and non-summer months, as well as among peak, semi-peak, and off-peak periods. Therefore, microgrid scheduling strategies should consider not only the total electricity purchased from the grid, but also the timing of grid electricity consumption. In addition, when the purchased power exceeds the contracted capacity or demand threshold, additional demand charges or over-contract penalties may be incurred, making BESS a useful tool for peak demand control. Related studies have pointed out that, in building microgrids or campus microgrids, jointly considering BESS scheduling with electricity price signals, demand-related charges, and battery usage costs can provide a more comprehensive evaluation of the economic value of energy storage systems in energy management [20][21].

In addition to electricity purchase costs and demand-related charges, renewable energy certificates may also affect the economic evaluation of microgrid energy management. Taking the TREC mechanism as an example, TREC can serve as proof of the environmental attributes of renewable electricity. Through certificate tracking, verification, and trading mechanisms, it supports corporate green electricity claims, Environmental, Social, and Governance performance improvement, and supply chain decarbonization needs [22][23]. Therefore, for microgrids equipped with PV generation systems, solar power generation can not only reduce external electricity purchases but may also generate additional revenue through renewable energy certificates. If TREC revenue is incorporated into the cost evaluation, the microgrid energy management problem involves not only electricity purchase costs, peak demand-related costs, and battery degradation costs, but also the certificate revenue generated by renewable energy production. This allows the overall operational economics of PV-BESS microgrids to be more comprehensively reflected.

Table 2.1 Taiwan TOU tariff

Classification						**Summer (May 16 – Oct 15)**	**Non-Summer**
Demand Charge	Regular Contracted Power				Per kW Per month	223.60	166.90
Flowing Electricity Prices	Monday to Friday	Peak	Summer	16:00~22:00	Per kWh	8.05	
		Half-Peak	Summer	09:00~16:00 22:00~24:00		5.02	
			Non-Summer	06:00~11:00 14:00~24:00			4.70
		Off-Peak	Summer	00:00~09:00		2.18	
			Non-Summer	00:00~06:00 11:00~14:00			2.00
	Saturday	Half-Peak	Summer	09:00~24:00		2.27	
			Non-Summer	06:00~11:00 14:00~24:00			2.10
		Off-Peak	Summer	00:00~09:00		2.18	
			Non-Summer	00:00~06:00 11:00~14:00			2.00
	Sunday	Off-Peak		All day		2.18	2.00

Based on the above characteristics, BESS scheduling in microgrids can generally be regarded as a multi-period decision-making problem. Scheduling methods need to determine the charging, discharging, or idle actions of the battery at different time points while reducing electricity purchase costs, peak demand-related costs, and battery usage costs under power balance constraints, SOC limits, and battery operating constraints. However, the effectiveness of BESS scheduling is affected by variations in load demand, PV generation, and electricity price signals. Moreover, the battery energy state is temporally coupled, meaning that the current decision affects the feasible operating space in subsequent time periods. Therefore, microgrid BESS scheduling is not merely a single-period cost minimization problem, but a sequential decision-making problem that must consider temporal dependency, operational constraints, and uncertain operating conditions. Recent studies have further incorporated battery degradation into microgrid scheduling to more accurately reflect the long-term economic impact caused by BESS operation [24][25]. This also highlights the necessity of developing scheduling methods that can maintain stable performance under real uncertain operating conditions.

2.2 Forecast Uncertainty in Microgrid Scheduling

Microgrid scheduling decisions are highly affected by variations in system operating conditions. In microgrids, PV generation and load demand are the main exogenous variables that influence power balance, grid purchase, BESS charging and discharging decisions, and system operating cost. However, both variables are subject to significant uncertainty. PV generation is affected by weather conditions, solar irradiance, cloud movement, and seasonal factors, while load demand is influenced by occupancy, user behavior, temperature changes, and daily operation patterns. Therefore, even when historical data and forecasting models are available, forecast errors remain difficult to eliminate in practical microgrid operation [26]-[28].

In microgrid energy management, many scheduling methods use future load demand, renewable generation, or electricity price information as decision inputs. With forecast information, the scheduling model can plan BESS charging and discharging actions in advance, such as reserving sufficient battery energy before expected high-price or high-load periods, or arranging charging when high PV generation is expected. However, when forecasted values deviate from actual operating conditions, the scheduling decisions originally determined based on forecast information may no longer match the actual system state, which may lead to increased cost or reduced operational efficiency [29][30].

Forecast uncertainty can affect BESS scheduling results in several ways. First, if future PV generation is overestimated, the scheduling model may postpone charging or reserve insufficient battery energy, causing additional electricity purchases from the grid when actual PV generation is lower than expected. Second, if future load demand is underestimated, the BESS may fail to preserve sufficient available energy before peak periods, thereby reducing the effectiveness of peak shaving and potentially increasing demand-related costs. Third, inaccurate forecasts may result in unnecessary or mistimed battery charging and discharging actions, increasing battery cycling without effectively reducing operating cost. These effects indicate that forecast errors not only affect the power balance of a single time period, but may also influence subsequent scheduling decisions through SOC dynamics and intertemporal energy constraints [30]-[32].

In addition, the impact of forecast errors on scheduling performance depends on how a scheduling method uses future information and how frequently its decisions are updated. If a scheduling method determines a complete schedule based on a fixed forecast horizon, deviations between forecasted data and actual operating conditions may cause the original schedule to continuously deviate from actual system requirements in subsequent periods. In contrast, methods that update decisions over time can incorporate the latest system state and revised forecast information, thereby reducing part of the impact caused by forecast errors. However, as long as the scheduling decision relies on future load demand or PV generation information, its final performance may still be limited by forecast quality. Therefore, forecast uncertainty is a common challenge that must be addressed in microgrid energy management [28][30][33].

Although improving forecasting models can reduce forecast errors, uncertainty in load demand and renewable generation cannot be eliminated in real-world microgrid operation. Therefore, scheduling methods that rely solely on a single deterministic forecast may experience performance degradation when actual operating conditions deviate from predicted values [30][31]. On the other hand, BESS scheduling itself has a clear intertemporal decision-making characteristic, where current charging and discharging actions affect the feasible operating space and cost performance in subsequent periods through SOC changes [34]. Therefore, in addition to improving forecasting models, developing scheduling methods that can make sequential decisions based on system states is also an important research direction in microgrid energy management. Based on this motivation, the next section further discusses the application of RL to microgrid BESS scheduling problems.

2.3 Reinforcement Learning for Microgrid Scheduling

Since microgrid BESS scheduling has a clear intertemporal decision-making characteristic, RL has been increasingly applied to microgrid energy management problems. Compared with traditional optimization methods that require explicit mathematical models to be formulated and solved at each scheduling stage, RL can formulate the energy management problem as a Markov decision process, where an agent selects control actions based on the current system state and learns a scheduling policy through long-term cumulative rewards. For BESS scheduling, the current charging and discharging decision not only affects the immediate electricity purchase cost, peak demand, and battery usage cost, but also influences the feasible operating space in subsequent periods through SOC changes. Therefore, the sequential decision-making nature of RL makes it suitable for handling time-dependent control problems in microgrid scheduling [35][36].

Early studies on microgrid energy management have applied Q-learning and its extensions to dispatch energy storage systems and distributed energy resources. However, traditional value-based RL methods are more suitable for discrete action spaces. When directly applied to BESS charging and discharging scheduling, the charging and discharging power usually needs to be discretized. Although this approach can simplify the learning problem, it may reduce control precision and lead to action space expansion when the action dimension increases. Since BESS charging and discharging power is inherently a continuous control variable, recent studies have increasingly adopted deep RL methods for microgrid scheduling problems to improve decision-making capability in high-dimensional state spaces and continuous action spaces [37][38].

Among RL methods, the actor-critic architecture has been widely applied to continuous control problems. This type of method typically consists of both a policy network and a value function network. The actor generates control actions according to the system state, while the critic evaluates the impact of the selected action on long-term cost or reward. For BESS scheduling, actor-critic methods can directly output continuous charging and discharging decisions, avoiding excessive discretization of battery power. At the same time, the critic can learn the long-term impact of current actions on future SOC, electricity purchase cost, and peak demand. Therefore, deep RL algorithms such as

DDPG, TD3, PPO, and SAC have been increasingly applied to microgrid energy management, building energy management, and energy storage scheduling problems [39][40].

Among these algorithms, SAC has attracted increasing attention in energy management and continuous control problems. SAC is an off-policy actor-critic method that incorporates the concept of maximum entropy RL, enabling the agent to maximize expected cumulative reward while maintaining a certain level of policy exploration [41]. This design helps improve exploration capability and training stability, making SAC suitable for continuous control problems such as BESS charging and discharging. Therefore, SAC has become one of the representative state-of-the-art RL algorithms for continuous control-based energy management problems, and has demonstrated good applicability in microgrid and building-level battery charging and discharging scheduling and energy management applications [42]-[44]. Recent studies have applied SAC to multi-timescale coordinated operation of microgrids, enabling the agent to generate corrective control actions based on real-time system states to address operational deviations caused by variations in load demand and renewable generation [45]. In addition, SAC has also been applied to energy storage scheduling problems to learn continuous charging and discharging control strategies and improve system operating cost performance [46]. Therefore, for microgrid BESS scheduling, SAC can serve as a suitable deep RL framework for handling continuous action spaces and long-term cost minimization problems.

However, standard RL methods may still be affected by the training data distribution and environmental transition uncertainty. In practical microgrid operation, even if the agent takes the same action under similar system states, the next-period load demand, PV generation, and system cost may still differ due to changes in external conditions. If the agent relies too heavily on a specific sample distribution during training, the learned policy may perform unstably when facing unseen operating conditions or conditions that deviate from the training distribution. Therefore, although RL provides an effective framework for solving BESS sequential scheduling problems, improving policy robustness under uncertain operating environments remains an important issue in microgrid energy management. This further motivates the discussion of robust energy scheduling and distributionally robust RL in the following section.

2.4 Robust Energy Scheduling

Microgrid operation involves various uncertainties, including load demand, PV generation, and system state transitions. In practical operation, system outcomes are affected not only by fluctuations in load demand and renewable generation, but also by factors that are difficult to fully model, such as detailed weather conditions, short-term solar irradiance variations, occupancy patterns, building equipment operating conditions, battery efficiency variations, measurement errors, and control delays. These factors may cause the same or similar system states and control actions to result in different subsequent outcomes. Therefore, energy scheduling methods should not only pursue the lowest operating cost, but also consider the robustness of decisions under uncertain operating environments. If a scheduling strategy relies excessively on a single forecast value or a specific historical data distribution, the original scheduling result may suffer from increased cost, reduced peak shaving performance, or lower battery utilization efficiency when actual operating conditions deviate from expectations. Therefore, developing stable scheduling decisions under uncertainty is an important research direction in microgrid energy management [28][47].

In traditional energy scheduling research, robust optimization is commonly used to address decision risks caused by uncertain parameters [48][49]. Such methods usually place uncertain factors, such as load demand, renewable generation, or equipment operating conditions, within a specific uncertainty set, and search for scheduling decisions that remain feasible and cost-stable under the worst-case scenario. Compared with deterministic scheduling that relies on a single forecast result, robust scheduling can reduce the sensitivity of scheduling outcomes to specific forecast errors or data shifts, thereby improving system reliability under uncertain operating conditions. However, an overly conservative uncertainty set may sacrifice part of the economic performance of the scheduling result, causing the system cost to be higher than the optimal solution under nominal conditions [50][51].

In addition to robust optimization, stochastic optimization is also commonly used in microgrid scheduling problems. Such methods usually use multiple possible scenarios to describe the probability distribution of uncertain variables, and make decisions based on minimizing the expected cost. Compared with robust optimization, which emphasizes feasibility and robustness under the worst-case scenario, stochastic scheduling focuses more on the average cost performance across different possible scenarios. However, stochastic optimization usually requires reliable probability distributions or scenario generation models. If the actual data distribution deviates from the assumed distribution, the scheduling result may still be affected. Therefore, robust and stochastic methods represent two major approaches to handling uncertainty, but they also face limitations related to conservativeness and distributional assumptions [29][52].

To balance worst-case robustness and probabilistic modeling, distributionally robust optimization has gradually been applied to energy scheduling and power system decision-making problems. Instead of assuming a single fixed probability distribution, distributionally robust optimization constructs a set of possible probability distributions and considers the decision performance under the worst-case distribution within this set. This approach can reduce the dependence on a single nominal distribution to some extent, while avoiding the conservativeness caused by focusing only on a single extreme scenario in traditional robust optimization. Therefore, distributionally robust optimization is suitable for handling energy scheduling problems involving distributional uncertainty in load demand, renewable generation, and market environments [53][54].

In the RL framework, uncertainty arises not only from forecast errors in exogenous variables, but also from variations in state transitions. For microgrid BESS scheduling, the same state-action pair may lead to different next states and subsequent costs due to differences in weather conditions, load behavior, building equipment operation, or battery states. If standard RL updates the value function only based on nominal transition samples in the training data, the critic may produce overly optimistic or overly distribution-dependent estimates, which can affect the stability of the learned policy under unseen operating conditions [55][56].

Therefore, robust RL and distributionally robust RL have been proposed to improve policy robustness under environmental uncertainty. These methods usually incorporate model uncertainty or transition distribution uncertainty into policy evaluation or critic updates, so that the agent learns not only a policy that performs well under average conditions, but also one that maintains relatively stable long-term rewards under state transitions that may deviate from the training distribution. For microgrid BESS scheduling, this design helps reduce the dependence of the scheduling policy on a specific data distribution or a single transition sample, thereby improving the robustness of the scheduling policy under variations in load demand and PV generation [8][55].

However, applying distributionally robust RL to continuous state and continuous action spaces is not straightforward, because robust policy evaluation needs to handle possible transition distribution shifts, and solving the worst-case transition model for each state-action pair can introduce a high computational burden. Related DR-SAC studies incorporate a distributionally robust formulation into the SAC framework and use generative modeling to approximate the nominal transition distribution, allowing distributionally robust RL to be applied more effectively to continuous control problems [8].

Based on the above discussion, the main purpose of robust energy scheduling is not merely to minimize cost under nominal forecast conditions, but to balance cost performance and stability under uncertain operating environments. For the PV-BESS microgrid scheduling problem investigated in this study, standard SAC provides sequential decision-making capability in a continuous action space, while the distributionally robust formulation further strengthens the critic's ability to handle transition uncertainty. Therefore, incorporating the distributionally robust concept into the SAC framework helps establish a BESS scheduling method that is more suitable for uncertain microgrid operating conditions.

2.5 Battery Degradation

In microgrid energy management, BESS can provide functions such as energy time-shifting, peak shaving, and renewable energy integration through charging and discharging operations. However, frequent operation also causes battery lifetime degradation. If the scheduling model only considers electricity purchase cost, self-consumption improvement, or peak demand control while ignoring the long-term cost caused by battery degradation, it may lead to overly frequent or aggressive charging and discharging actions. As a result, short-term operating cost may be reduced at the expense of increased battery lifetime loss. Therefore, incorporating battery degradation cost into BESS scheduling and microgrid energy management is an important step for evaluating the true economic value of energy storage systems [57][58].

Battery degradation is generally divided into calendar aging and cycle aging. Calendar aging refers to the natural aging of a battery over time, even when no charging or discharging operation is performed, and is affected by factors such as time, temperature, and state of charge. Cycle aging is mainly associated with actual battery usage and is influenced by the number of charge-discharge cycles, depth of discharge, energy throughput, charge-discharge rate, and operating temperature. For BESS scheduling problems, calendar aging reflects the depreciation cost of the battery as an asset over time, while cycle aging reflects the operational lifetime loss caused by scheduling decisions. Therefore, to reasonably evaluate the long-term economics of BESS, both types of degradation factors should be included in the cost analysis [59][60].

Existing studies handle battery degradation in different ways. Some studies consider degradation as an evaluation metric of scheduling results rather than directly incorporating it into the optimization objective. In this type of setting, the model usually first generates a dispatch schedule based on cost, self-consumption, or other operational objectives, and then uses an empirical aging model or degradation assessment to analyze the impact of different scheduling strategies on battery lifetime. This approach helps evaluate the aging impact caused by different operating strategies and also indicates that battery degradation is an important factor in assessing the long-term benefits of PV-BESS scheduling [59][61].

In addition, some studies consider only either calendar lifetime or cycle lifetime when modeling battery lifetime. Methods that use only calendar lifetime usually treat battery lifetime as a fixed replacement period, which can reflect time-related asset depreciation. Methods that use only cycle lifetime can reflect the impact of battery usage frequency and energy throughput on lifetime. However, calendric aging and cyclic aging occur simultaneously, and the actual battery lifetime may be jointly constrained by both. Therefore, if the scheduling model can consider both calendar degradation and cycle degradation, it can more comprehensively describe the impact of BESS operation on long-term cost [58][60].

For microgrid BESS scheduling, ignoring battery degradation may cause the scheduling strategy to overuse the battery in pursuit of short-term reductions in electricity purchase cost or peak demand cost. However, when degradation cost is incorporated into the objective function, the scheduling strategy needs to balance electricity cost savings and battery lifetime loss. For example, charging during low-price periods and discharging during high-price periods can provide energy arbitrage benefits, but if the price difference is insufficient to compensate for battery degradation cost, frequent charging and discharging may instead reduce the overall economic benefit. Therefore, degradation-aware scheduling can make BESS operation more consistent with the goal of long-term cost minimization [62][63].

Based on the above reasons, this study incorporates battery degradation cost into the total operating cost to avoid unreasonable charging and discharging behavior caused by ignoring battery lifetime loss. Since this study mainly compares the cost performance and operational characteristics of different BESS scheduling methods under the same system settings, battery degradation cost is divided into calendar degradation cost and cycle degradation cost, and both are converted into daily costs and included in the total operating cost calculation. This setting allows MILP, SAC, and DR-SAC to be compared under a consistent cost framework and provides a more complete evaluation of the long-term economic performance of PV-BESS microgrid scheduling strategies.

2.6 Research Gap

Based on the literature reviewed above, microgrid energy management and BESS scheduling have been widely applied to renewable energy integration, electricity cost reduction, peak demand control, and self-consumption improvement. Existing studies have also discussed the impacts of forecast uncertainty, robust scheduling, RL, and battery degradation cost on scheduling performance. These studies indicate that PV-BESS microgrid scheduling is not merely a single-period cost minimization problem, but a sequential decision-making problem involving intertemporal energy states, variations in external operating conditions, battery lifetime degradation, and multiple cost components.

However, in practical microgrid operation, PV generation and load demand remain difficult to forecast with complete accuracy. When scheduling methods heavily rely on future forecast information, deviations between actual operating conditions and forecasted values may cause the originally determined charging and discharging strategy to no longer match the actual system state. This may further affect electricity purchase cost, peak demand control, and battery utilization efficiency. Although traditional robust optimization and stochastic optimization have been used to address uncertainty, the former may be affected by the design and conservativeness of the uncertainty set, while the latter depends on probability distributions or scenario generation quality. Therefore, maintaining cost performance and scheduling stability under uncertain operating conditions remains an important issue in PV-BESS microgrid scheduling [64][65].

In this context, RL provides a suitable framework for handling BESS sequential decision-making problems. In particular, actor-critic methods such as SAC can handle continuous charging and discharging power decisions, making them suitable for BESS scheduling. However, standard RL methods usually update the value function based on nominal transition samples from the training data. If the actual load demand, PV generation, or system state transition during operation differs from the training data distribution, the learned policy may still exhibit unstable performance. Therefore, using standard RL alone may not be sufficient to fully address transition uncertainty in microgrid operation, which further highlights the need to incorporate the distributionally robust concept into RL-based scheduling.

In addition, the economic benefits of BESS scheduling should not be evaluated only by electricity purchase cost or peak demand cost. Frequent battery charging and discharging cause lifetime degradation. If battery degradation cost is ignored, the scheduling strategy may tend to overuse the battery, leading to short-term cost reduction while underestimating long-term economic impacts. Some studies have started to incorporate battery aging into the analysis, but degradation may still be treated as a post-scheduling evaluation metric or only partially considered. Therefore, when comparing different BESS scheduling methods, incorporating both calendar degradation cost and cycle degradation cost into a consistent total cost framework can help more comprehensively reflect the impact of BESS operation on long-term operating cost.

Based on the above research gaps, this study investigates a BESS scheduling method that combines SAC with a distributionally robust formulation for PV-BESS microgrid scheduling. The main focus is to enable the agent to make battery charging and discharging decisions in a continuous action space, while improving the robustness of the policy against transition uncertainty through distributionally robust critic updates. For cost evaluation, this study further incorporates electricity purchase cost, contract capacity-related cost, battery degradation cost, and TREC revenue to establish a more comprehensive total operating cost framework for PV-BESS microgrids. By comparing SAC, DR-SAC, and the theoretical optimum, this study aims to evaluate the impact of distributionally robust RL on the cost performance and policy stability of BESS scheduling under uncertain operating conditions.

CHAPTER 3
METHODOLOGY

3.1 System Framework and Problem Definition

This study investigates the battery charging and discharging scheduling problem for a grid-connected PV-BESS microgrid. The system primarily consists of PV generation, building load, a BESS, and the utility grid. At each time step, decisions are made regarding how power flows among the PV system, the grid, the BESS, and the building load, to satisfy the building's electricity demand while minimizing the overall system operating cost.

The objective of this study is to minimize the total operating cost of the microgrid over the testing period. Cost components include electricity purchase cost, battery degradation cost, and contract capacity-related cost. In addition, revenue from TREC is incorporated as a negative cost term in the total operating cost calculation. To ensure consistency across different methods, this study applies the same system configuration, cost structure, and operational constraints to MILP, SAC, and DR-SAC.

The overall methodology consists of two parts. First, a MILP model is formulated, which assumes complete future information to derive a theoretical optimum serving as the benchmark for evaluating RL-based methods. Second, RL-based scheduling models are developed, enabling SAC and DR-SAC to make sequential decisions based on the current system state without relying on complete future information. The following subsections describe the microgrid system configuration and the overall research framework of this study.

3.1.1 Microgrid System Configuration

The microgrid system considered in this study is illustrated in Fig. 3.1, and primarily comprises a PV generation system, building load, a BESS, and the utility grid. PV generation can directly supply the building load or be used to charge the BESS. The utility grid can supply the building load and, when necessary, charge the BESS. The

BESS can discharge during periods of high electricity prices or high grid demand to support the building load.

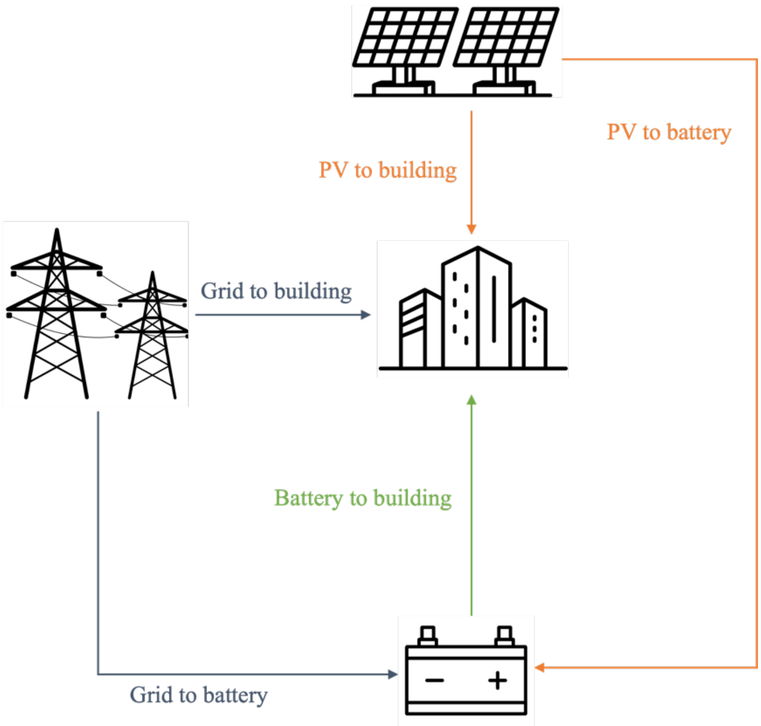

Fig. 3.1 Energy flow between the microgrid

Based on the energy flow depicted in Fig. 3.1, the power supply sources in the system include the power supplied from PV to the building, the power discharged from the BESS to the building, the power supplied from the grid to the building, the power supplied from PV to the BESS, and the power supplied from the grid to the BESS. The electricity demand side includes the building load demand and the BESS charging power. Accordingly, the overall power balance relationship can be expressed as follows:

$$P_{PV2B}(t) + P_{dch}(t) + P_{G2B}(t) + P_{PV2BESS}(t) + P_{G2BESS}(t)$$
$$= P_{load}(t) + P_{ch}(t) \quad (1)$$

where $P_{PV2B}(t)$ denotes the power supplied from PV to the building load at time t, $P_{dch}(t)$ denotes the power discharged from the BESS to the building load at time t , $P_{G2B}(t)$ denotes the power supplied from the utility grid to the building load, $P_{PV2BESS}(t)$ denotes the charging power supplied from PV to the BESS, $P_{G2BESS}(t)$

denotes the charging power supplied from the utility grid to the BESS, $P_{load}(t)$ denotes the building load demand, and $P_{ch}(t)$ denotes the total charging power of the BESS.

Based on the aforementioned system configuration, the proposed cost model simultaneously considers electricity purchase costs under the TOU pricing mechanism and contract capacity-related costs to evaluate the effectiveness of BESS scheduling in reducing electricity purchase costs and controlling peak demand.

3.1.2 Research Framework

This study's methodological design is based on the same microgrid scheduling problem, in which different scheduling methods are evaluated under consistent system settings, testing data, and cost calculation framework. Historical data on building load demand, PV generation, and TOU electricity prices are used to characterize the operating conditions of the microgrid at different time periods. These data serve as the common data source for both the mathematical programming model and the RL environment.

In the mathematical programming component, a MILP model is formulated to describe system costs, power balance, battery charging and discharging constraints, and contract capacity costs. Based on this mathematical model, a theoretical optimum can be obtained under the assumption of complete future information, serving as the optimal benchmark under the same system configuration. This theoretical optimum is not intended as a directly deployable scheduling method, but rather as a benchmark for evaluating the gap between RL-based scheduling results and the best achievable cost performance. In addition, building upon the same mathematical framework, a more practically oriented MILP scheduling method is incorporated, which performs scheduling under limited forecast information and serves as a reference for analyzing the impact of forecast limitations on total operating cost.

In the RL component, the microgrid scheduling problem is formulated as a sequential decision-making problem. At each time step, the agent determines the charging and discharging behavior of the BESS based on the current system state and updates its policy through cost-based feedback. SAC serves as the baseline RL method, while DR-SAC further incorporates the concept of distributional robustness [8] into the SAC framework, enabling the policy to reduce dependence on specific training data distributions and improve the stability of scheduling decisions when facing variations in load demand, PV generation, and operating conditions.

Therefore, the main comparison in this study focuses on the cost performance and operational characteristics of SAC, DR-SAC, and the theoretical optimum to evaluate whether DR-SAC can achieve scheduling results closer to the optimal benchmark than SAC and demonstrate more robust scheduling capability under environmental uncertainty. The practical MILP scheduling method is included as a supplementary comparison to examine how mathematical programming-based scheduling is affected by limited forecast information. The primary evaluation metric is total operating cost, supplemented by individual cost components, including electricity purchase cost, battery degradation cost, contract capacity-related cost, and TREC revenue.

3.2 Mathematical Formulation

This section formulates the mathematical model for the microgrid battery charging and discharging scheduling problem, describing the power flows, cost components, battery degradation model, and operational constraints within the system. The primary purpose of this mathematical model is to explicitly define the physical relationships and cost calculation logic of the microgrid system, so that different scheduling methods can be compared under consistent system conditions.

In MILP-based scheduling methods, the power flow variables, battery state variables, and contract capacity-related variables defined in this section are directly used as decision variables in the optimization model, and scheduling results are obtained by solving the objective function subject to the defined constraints. In contrast, in RL-based scheduling, the agent does not directly optimize the complete set of mathematical programming variables, but instead outputs battery charging and discharging decisions based on the current system state. These decisions are then translated into actual scheduling outcomes through the same system power balance, battery state update, cost calculation, and operational constraints.

Therefore, the mathematical model established in this section does not imply that all methods adopt the same form of decision variables. Rather, it provides a unified system description and evaluation benchmark. MILP-based methods use this model for optimization, while RL-based methods use it to construct the environment dynamics, reward function, and cost evaluation during the testing phase. Through this design, it is ensured that the theoretical optimum, the practical MILP scheduling method, SAC, and DR-SAC are all compared under the same system configuration, cost structure, and physical constraints, thereby avoiding unfair evaluation results caused by differing model assumptions.

The remainder of this section sequentially describes the decision variables and parameters, the objective function, the battery degradation model, and the system operational constraints.

3.2.1 Decision Variables and Parameters

This section presents the main decision variables and parameters used in the mathematical model, which serve as the basis for formulating the subsequent objective function and system constraints. The overall optimization problem is formulated over a 24-hour scheduling horizon and discretized on an hourly basis; therefore, each time step corresponds to one hour.

The decision variables used in this study are summarized in Table 3.1 and are primarily used to describe the power flows among system components as well as the charging and discharging behavior of the BESS. The optimization model solves for the optimal value of each decision variable at every time step to achieve the objective of minimizing total operating cost.

In addition, binary variables are introduced to represent the operating state of the BESS, distinguishing between charging and discharging behavior and preventing both from occurring simultaneously.

The parameters used in this study are summarized in Table 3.2 and include electricity tariff-related parameters, battery characteristic parameters, and system operation-related parameters such as PV generation output, load demand, and energy storage system capacity.

Table 3.1 Overview of decision variables used in the mathematical model

Representation	Variables	Unit
P_{G2BESS}	Power from grid to BESS	kW
P_{G2B}	Power from grid to building	kW
$P_{PV2BESS}$	Power from PV to BESS	kW
P_{PV2B}	Power from PV to building	kW
$P_{curtail}$	PV power curtailment	kW
P_{BESS2B}	Power from BESS to building	kW
E_{BESS}	Stored energy of the BESS	kWh
P_{ch}	BESS charging power	kW
P_{dch}	BESS discharging power	kW
u_{ch}	BESS Charge state	Binary
u_{dch}	BESS Discharge state	Binary

Table 3.2 Overview of parameters used in the mathematical model

Representation	Variables	Unit
P_{load}	Load Demand	kW
P_{PV}	Power generated by PV	kW
Pr_{buy}	Grid purchasing price	NTD
Pr_{TREC}	Average TREC price per kWh	NTD/kWh
C_{Bat}^{Ins}	Unit installation cost of the BESS	NTD
C_{Bat}^{E}	Unit energy capacity cost of the BESS	NTD
C_{Bat}^{P}	Unit power cost of the BESS	NTD
L_{cal}	Calendar lifetime of the battery	Years
L_{cyc}	Cycle lifetime of the battery	Cycles
$E_{BESS,cap}$	Installed energy capacity of the BESS	kWh
$P_{BESS,max}$	Rated power of the BESS	kW
η_{ch}	Battery charging efficiency	-
η_{dch}	Battery discharging efficiency	-

P_{CP}	Contract capacity	kW
λ_m	Unit price of contract capacity	NTD/kW

3.2.2 Objective Function

As shown in Eq. (2), the objective is to minimize the total daily operating cost of the microgrid. The cost is calculated on a daily basis, and the objective function consists of the sum of hourly electricity purchase costs over 24 hours $\sum_{t=1}^{24} C_{Energy}\ (t)$, the daily battery degradation cost C_{Deg}^{day}, and the daily contract capacity cost C_{CP}^{day}, minus the daily TREC revenue Rev_{TREC}^{day}.

$$\min C_{total}^{day} \tag{2}$$

$$C_{day} = \sum_{t=1}^{24} C_{Energy}\ [t] + C_{Deg}^{day} + C_{CP}^{day} - R^{day} \tag{3}$$

Regarding electricity purchase cost, the system purchases electricity from the grid according to the TOU tariff at each time step, as shown in Eqs. (4) and (5). The electricity purchase cost is calculated based on the electricity price at each time step and the grid purchase power during that hour, where the purchased power includes power supplied to the building load and power used for battery charging.

$$C_{Energy}(t) = Pr_{buy}(t) \times P_{buy}(t) \tag{4}$$

$$P_{buy}(t) = P_{G2BESS}(t) + P_{G2B}(t) \tag{5}$$

Battery degradation cost is used to measure the lifetime loss incurred by the battery during operation. Battery aging can be classified into time-related calendar aging and usage-related cycle aging [60][61], and the battery's actual useful life is governed by whichever of the two lifetimes is reached first [60][66][67]. As shown in Eqs. (6) and (7), this study considers the effects of both types of aging and converts them into daily costs to be incorporated into the overall optimization objective. The detailed calculation method is described in Section 3.2.3.

$$C_{Deg}^{day} \geq C_{deg,cal}^{day} \tag{6}$$

$$C_{Deg}^{day} \geq C_{deg,cyc}^{day} \tag{7}$$

Contract capacity cost refers to the basic charge that the system pays based on the pre-registered contract capacity. When the actual power consumption exceeds the contract capacity, additional penalty charges are incurred, calculated using a tiered pricing scheme based on the degree of exceedance. As shown in Eq. (8), this cost consists of a basic charge and a two-tier exceedance penalty, where P_{CP} is the contract capacity, λ_m is the basic electricity rate, and P_{ex1} and P_{ex2} are the exceedance power in the first and second tiers, respectively.

$$C_{CP} = (P_{CP} \times \lambda_m) + (P_{ex1} \times \lambda_m \times 2) + (P_{ex2} \times \lambda_m \times 3) \tag{8}$$

Furthermore, the renewable energy generated by the PV system can correspond to the issuance of TREC, with revenue proportional to the amount of electricity generated. As shown in Eq. (9), this revenue is represented as a deduction term in the objective function, calculated as the product of the certificate price Pr_{TREC} and the total daily PV generation, reflecting its cost-offsetting effect on total operating costs. The average certificate price is 4.63 NTD/kWh [68].

$$R^{day} = Pr_{TREC} \times \left(\sum_{t=1}^{24} P_{PV}(t) \right) \tag{9}$$

Regarding the composition and allocation of PV generation, as shown in Eq. (10), the total generation $P_{PV}(t)$ at each time step consists of three components: power supplied to the BESS $P_{PV2BESS}(t)$, power directly supplied to the building load $P_{PV2B}(t)$, and curtailed power $P_{curtail}(t)$ arising from system operational constraints or storage saturation. Through this power balance relationship, the allocation, and losses of PV generation within the microgrid system are fully accounted for.

$$P_{PV}(t) = P_{PV2BESS}(t) + P_{PV2B}(t) + P_{curtail}(t) \tag{10}$$

In summary, the objective function of this study integrates electricity purchase cost, battery degradation cost, contract capacity cost, and renewable energy revenue to obtain the minimum total operating cost subject to system operational constraints.

3.2.3 Battery Degradation Model

Accurately quantifying battery capacity degradation costs is critical when making BESS charging and discharging scheduling decisions and evaluating long-term economic benefits. The battery degradation model constructed in this study primarily consists of two components: calendar degradation and cycle degradation.

First, referring to the cost calculation method in [69], the total investment cost of the battery system C_{Bat} is defined as the basis for subsequent calculations. This cost integrates the unit energy capacity cost C_{Bat}^{E}, the unit installation cost C_{Bat}^{Ins}, and the unit power cost C_{Bat}^{P}, as expressed in Eq. (11). The corresponding parameter values used in this equation are summarized in Table 3.3.

$$C_{Bat} = (C_{Bat}^{E} + C_{Bat}^{Ins})E_{BESS,cap} + C_{Bat}^{P}P_{BESS,max} \tag{11}$$

Table 3.3 Economic and Investment Parameters of the BESS

BESS cost and lifetime parameters	
Unit energy capacity cost	19200 NTD/kWh
Unit installation cost	480.0 NTD/kWh
Unit power cost	11520.0 NTD/kW
Calendar lifetime	20 years
Cycle lifetime	6000 cycles

Based on the total investment cost, the daily calendar degradation cost $C_{deg,cal}^{day}$ describes the depreciation of the battery due to the natural passage of time. Following the annual replacement cost evaluation method proposed in [60], this study adapts the logic into a daily basis by evenly distributing the total battery cost over the total calendar lifetime L_{cal} of the battery, as shown in Eq. (12).

$$C_{deg,cal}^{day} = \frac{C_{Bat}}{L_{cal} \cdot 365} \tag{12}$$

On the other hand, cycle degradation cost reflects the physical wear caused by actual charging and discharging operations. The daily cycle degradation cost $C_{deg,cyc}^{day}$ is obtained by summing the product of the energy throughput and the unit cycle cost c_{cyc} over 24 hours, as expressed in Eq. (13).

$$C_{deg,cyc}^{day} = \sum_{t=1}^{24} c_{cyc} E_{th}(t) \tag{13}$$

where the unit cycle cost c_{cyc} represents the asset depreciation per unit of energy throughput, as defined in Eq. (14).

$$c_{cyc} = \frac{C_{Bat}}{2 L_{cyc} E_{BESS,cap}} \tag{14}$$

Here, L_{cyc} is the cycle lifetime of the battery at a specific depth of discharge, and the factor 2 in the denominator represents that one complete cycle consists of both a charging and a discharging process.

To accurately compute the hourly energy throughput $E_{th}(t)$, the model incorporates the charging efficiency η_{ch} and discharging efficiency η_{dch} to reflect energy losses during the conversion process, as defined in Eq. (15).

$$E_{th}(t) = \left(\eta_{ch} P_{ch}(t) + \frac{P_{dch}(t)}{\eta_{dch}} \right) \tag{15}$$

In the system operation logic, the BESS charging power consists of electricity from both PV and the grid, as shown in Eq. (16), while the BESS discharging power is used to serve the load demand, as shown in Eq. (17).

$$P_{ch}(t) = P_{PV2BESS}(t) + P_{G2BESS}(t) \tag{16}$$

$$P_{dch}(t) = P_{BESS2B}(t) \tag{17}$$

Through the above model, the physical aging process of the battery is translated into concrete economic costs, which together with the electricity purchase costs form the overall optimization objective, ensuring that the energy storage system achieves a balance between economic efficiency and longevity.

3.2.4 System Constraints

To describe the various operational limitations of the system, constraints are imposed on battery operating behavior, power flows, and overall operational boundaries.

The charging and discharging behavior of the BESS is subject to rated power and operating state constraints. Eqs. (18) and (19) limit the charging and discharging power to not exceed the maximum rated power of the battery, while Eqs. (20) and (21) use binary variables to ensure that the battery can only either charge or discharge at any given time, but not both simultaneously.

$$0 \leq P_{ch}(t) \leq P_{BESS,max} \cdot u_{ch}(t) \tag{18}$$

$$0 \leq P_{dch}(t) \leq P_{BESS,max} \cdot u_{dch}(t) \tag{19}$$

$$u_{ch}(t) + u_{dch}(t) \leq 1 \tag{20}$$

$$u_{ch}(t), u_{dch}(\mathrm{t}) \in \{0,1\} \tag{21}$$

To ensure that all power variables are physically meaningful, Eqs. (22)– (26) impose non-negativity constraints on all energy flows.

$$P_{G2BESS}(\mathrm{t}) \geq 0 \tag{22}$$

$$P_{G2B}(\mathrm{t}) \geq 0 \tag{23}$$

$$P_{PV2BESS}(\mathrm{t}) \geq 0 \tag{24}$$

$$P_{PV2B}(\mathrm{t}) \geq 0 \tag{25}$$

$$P_{curtail}(\mathrm{t}) \geq 0 \tag{26}$$

The state of charge (SOC) of the battery is constrained within a certain range to prevent overcharging or over-discharging, as specified in Eq. (27).

$$0.2 \leq SOC_{BESS}(\mathrm{t}) \leq 0.9 \tag{27}$$

The change in stored energy of the BESS over time is described by an energy balance equation that accounts for charging and discharging efficiency, as shown in Eq. (28). The corresponding relationship between SOC and stored energy is defined by Eq. (29).

$$E_{BESS}(\mathrm{t}) = E_{BESS}(\mathrm{t}-1) + \left[\eta_{ch} \cdot \left(P_{ch}(\mathrm{t})\right) - \frac{P_{dch}(\mathrm{t})}{\eta_{dch}}\right] \cdot \Delta t \tag{28}$$

$$SOC_{BESS}(\mathrm{t}) = \frac{E_{BESS}(\mathrm{t})}{E_{BESS,cap}} \tag{29}$$

To maintain operational continuity across days, Eq. (30) constrains the initial battery energy of each day to equal the final energy of the previous day.

$$E_{BESS}(0)_d = E_{BESS}(24)_{d-1} \tag{30}$$

Finally, contract capacity-related constraints are described by Eqs. (31)– (34), where the grid supply power must not exceed the maximum allowable value, and additional variables are introduced to represent the range of capacity exceedance.

$$P_{G2BESS}(t) + P_{G2B}(t) \leq \mathrm{P_{buy}}(\mathrm{t}) \tag{31}$$

$$P_{buy}(\mathrm{t}) \leq P_{CP} + P_{ex1} + P_{ex2} \tag{32}$$

$$0 \leq P_{ex1} \leq 0.1 \times P_{CP} \tag{33}$$

$$0 \leq P_{ex2} \tag{34}$$

3.3 MILP-Based Theoretical Optimum

Building upon the mathematical model established in Section 3.2, this section further develops a theoretical optimum model to serve as an idealized cost optimization benchmark. This theoretical optimum adopts the same cost structure, power balance relationships, battery degradation model, and system operational constraints as described in the preceding mathematical model. The key difference is that the theoretical optimum assumes all exogenous data over the testing period to be known in advance, including building load demand, PV generation, and electricity prices at each time step. Therefore, the MILP model can be solved to optimality over the entire testing period under the assumption of complete future information.

The testing period in this study comprises 50 days in total, with each day scheduled on a 24-hour basis. Accordingly, the objective function of the MILP-based theoretical optimum can be expressed as follows:

$$\min C_{Total} \tag{35}$$

$$C_{Total} = \sum_{d=1}^{50}\left(\sum_{t=1}^{24} C_{Energy}\,(d,t) + C_{Deg}^{day}(d) + C_{CP}^{day}(d) - R_{TREC}^{day}(d)\right) \tag{36}$$

where $d \in \{1, \dots, 50\}$ indexes the day within the testing period and $t \in \{1, \dots, 24\}$ indexes the hour within each day. The definitions of each cost component are consistent with those in Section 3.2, with the difference that daily costs are here aggregated over the 50-day testing period to obtain the minimum total operating cost across the entire testing horizon.

In constructing the theoretical optimum, the building load demand, PV generation, and TOU electricity prices over the entire 50-day testing period are assumed to be fully available to the MILP model. Since these data are historical testing data rather than forecasted values, the model can optimize the battery charging and discharging schedule and all power flows over the entire testing horizon in a single solve. This setting is therefore equivalent to assuming complete future information from the perspective of real-time operation, and the obtained result represents the lowest achievable cost under the same system configuration, cost function, and operational constraints.

However, the theoretical optimum does not represent a scheduling strategy that can be directly implemented in real-time operation, as future load demand and PV generation cannot be fully known in practice. Its primary purpose is to provide an idealized optimization benchmark, against which the scheduling results of SAC and DR-SAC can be compared to assess the cost gap between the RL-based methods and the optimal solution.

3.4 DR-SAC Framework for BESS Scheduling

This study adopts SAC [41] as the baseline RL method and further employs DR-SAC [8] as a distributionally robust extension to learn continuous BESS charging and discharging policies in the microgrid. The mathematical formulation established in Section 3.2, including the microgrid power balance, BESS operating constraints, SOC transition, energy purchase cost, battery degradation cost, contract capacity cost, and TREC revenue, provides the basis for constructing the RL environment. Therefore, this section does not reformulate the complete physical and cost models. Instead, it focuses on how the microgrid operation model defined in Section 3.2 is incorporated into the SAC and DR-SAC scheduling frameworks.

In the DR-SAC-based scheduling framework, the agent observes the current microgrid state at each hourly time step and outputs a continuous action as the BESS charging or discharging decision. The environment then updates the BESS operation, SOC, grid purchase power, peak demand, and corresponding operating cost based on the selected action, and returns the computed reward and next state to the agent. Since the reward function is derived from the operating cost, the process of maximizing cumulative reward can be interpreted as minimizing the long-term total operating cost.

The key distinction between DR-SAC and standard SAC [41] lies in its incorporation of transition distribution uncertainty into the critic update process. Standard SAC typically estimates the next-state value based on observed transition samples and uses this estimate to update the Q-network. However, in practical microgrid operation, load demand, PV generation, and operating conditions are inherently uncertain, meaning that the same state-action pair may lead to different next-state outcomes. If the critic is updated only based on nominal transition samples, it may produce overly optimistic Q-value estimates under certain operating conditions, thereby affecting the stability of the learned policy.

To mitigate this issue, DR-SAC assumes that the true state transition distribution may deviate from the nominal transition distribution reflected by the training data, and it performs worst-case evaluation within an uncertainty set defined by Kullback–Leibler (KL) divergence [70]. Through this design, the critic update no longer relies only on a single observed next state, but instead considers possible next-state variations and adopts a more conservative robust critic target. This mechanism prevents the agent from making over-optimistic value estimations regarding uncertain environments, improving the robustness of the BESS scheduling policy under variations in load demand, PV generation, and operating conditions.

To realize the robust critic update in a continuous state space, this study introduces a Variational Autoencoder (VAE) as the transition model[71][72]. The VAE learns the latent distribution of state transitions from the replay buffer and, during training, generates multiple possible next-state samples based on the current state and action. It should be noted that the VAE is not used to provide operational forecasts of future load demand or PV generation. Instead, it serves as a generative model for approximating the nominal transition distribution in the critic update. These generated next-state samples are then substituted into the robust Bellman operator and combined with the dual variable β estimated by the G-network [73] to compute a conservative critic target that accounts for the worst-case future value. This target replaces the standard SAC approach of relying solely on the observed transition sample for critic updating.

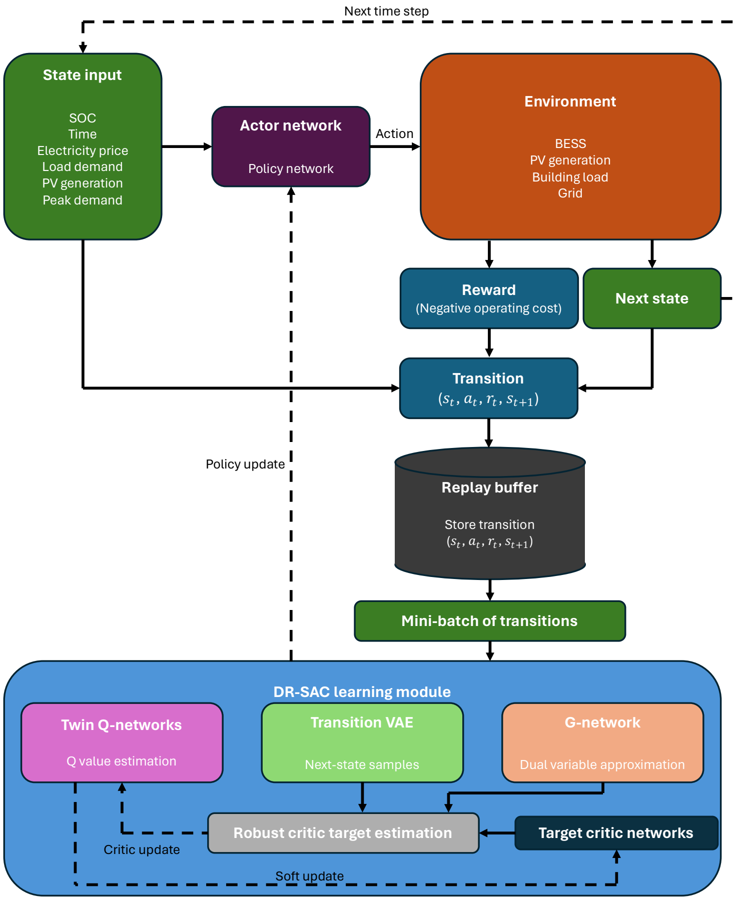

Fig. 3.2 DR-SAC Training Framework

The following sections describe the RL problem formulation, the DR formulation, the network architecture, and the training procedure. The overall methodological framework is illustrated in Fig. 3.2.

3.4.1 Reinforcement Learning Problem Formulation

This study formulates the BESS scheduling problem as a Markov Decision Process (MDP), enabling the agent to interact with the microgrid environment at each hourly time step. The MDP is defined as:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma) \tag{37}$$

where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ denotes the action space, P denotes the state transition dynamics, r denotes the reward function, and γ denotes the discount factor.

At each time step t, the state s_t observed by the agent consists of six features:

$$s_t = [SOC, hour, P_{peak}, Pr_{buy}, P_{load}, P_{PV}] \tag{38}$$

The state covers battery status, time information, peak demand, and current power system conditions. All elements represent information available at the beginning of time step t, and the agent selects action a_t without relying on future information.

Let $P_{BESS,t}$ denote the net charging and discharging power of the BESS, where a positive value indicates charging and a negative value indicates discharging. The agent outputs a continuous action $a_t \in [-1,1]$ at each time step, which is mapped to the actual BESS charging or discharging power:

$$P_{\text{BESS},t} = a_t \times P_{\text{BESS,max}} \tag{39}$$

where $P_{BESS,t} > 0$ indicates charging and $P_{BESS,t} < 0$ indicates discharging, corresponding to P_{ch} and P_{dch} in Section 3.2. The actual charging and discharging power is subject to the SOC upper and lower bounds, as detailed in Section 3.2.4.

After the action is executed, the environment updates the system state according to the microgrid operation model:

$$s_{t+1} = f(s_t, a_t) \tag{40}$$

This encompasses BESS charging and discharging power calculation, SOC update, power balance allocation, and monthly peak demand update. The detailed calculation logic is provided in Section 3.2.

The reward is calculated based on the total operating cost at each time step:

$$r_t = -C_t \tag{41}$$

where C_t is the total operating cost at time step t, corresponding to the cost components defined in Section 3.2.2, including energy purchase cost, battery degradation cost, contract capacity cost, and TREC revenue. The contract capacity cost is updated incrementally at each time step.

The DR-SAC agent aims to learn a policy π that maximizes the expected cumulative discounted reward over the scheduling horizon:

$$\max_{\pi} E_{\pi}\left[\sum_{t=1}^{T} \gamma^{t-1} r_t\right] \tag{42}$$

Since the reward is defined as the negative total operating cost, maximizing cumulative reward is equivalent to minimizing the long-term total operating cost of the microgrid. However, this formulation assumes fixed and known transition dynamics, without accounting for uncertainty in actual operation. Section 3.4.2 extends this to a DR formulation to improve policy robustness under environmental uncertainty.

3.4.2 Distributionally Robust Formulation

The key difference between DR-SAC and standard SAC lies in the computation of the critic target. DR-SAC incorporates transition distribution uncertainty into the SAC framework and considers the worst-case transition distribution within a KL-divergence constrained uncertainty set to improve the robustness of the learned policy under environmental uncertainty.

In the BESS scheduling problem, after the agent executes a charging or discharging decision at time t, the next system state may be affected by load demand fluctuations, PV generation uncertainty, SOC transitions, and peak demand variations. Therefore, the same state-action pair (s_t, a_t) may correspond to different next-state outcomes under different operating conditions. If the critic estimates future value solely based on the nominal transition, it may lead to excessive dependence on specific transition samples in Q-value estimation, thereby reducing the robustness of the learned scheduling policy under uncertain environments.

To characterize transition distribution uncertainty, let $p_0(\cdot \mid s_t, a_t)$ denote the nominal transition distribution reflected by the training data, and let p denote a possible perturbed transition distribution. Following the KL-divergence constrained uncertainty set in DR-SAC, the transition uncertainty set for state-action pair (s_t, a_t) is defined as:

$$U(s_t, a_t) = \{p: D_{KL}(p|p_0(\cdot\,|s_t, a_t)) \leq \delta\} \tag{43}$$

where δ is the robustness radius controlling the allowable deviation of the possible transition distribution from the nominal transition distribution. In this study, δ is set to 0.01 as specified in Table 3.4.

Based on this uncertainty set, the future value term in the SAC critic target is modified into a distributionally robust form. The resulting DR-SAC critic target is expressed as:

$$y_t^{DRSAC} = r_t + \gamma \min_{p \in U(s_t, a_t)} E_{s_{t+1} \sim p} \left[\min_{j=1,2} Q_{\overline{\theta_J}}(s_{t+1}, a_{t+1}) - \alpha \log \pi\,(a_{t+1}|s_{t+1}) \right] \tag{44}$$

where γ is the discount factor, α is the entropy temperature parameter, and $Q_{\overline{\theta_J}}$ denotes the target critic networks. The minimum over twin target critics $\min_{j=1,2} Q_{\overline{\theta_J}}$ is taken to reduce the risk of Q-value overestimation [74]. $a_{t+1} \sim \pi(\cdot\,|s_{t+1})$ denotes the action sampled by the policy network at the next state.

The $\min_{p \in U(s_t, a_t)}$ term in the above formula indicates that DR-SAC selects the worst-case transition distribution among all possible transition distributions that minimizes the expected future value. $E_{s_{t+1} \sim p}$ denotes the expected future value over possible next states under that transition distribution. In other words, DR-SAC does not perform critic updates based solely on a single observed next state, but instead constructs a more conservative critic target through the worst-case expected future value.

In practice, the nominal transition distribution $p_0(\cdot\,|s_t, a_t)$ of the microgrid environment has no closed-form expression. Therefore, the transition VAE learns the state transition relationship between (s_t, a_t) and s_{t+1} from transition samples in the replay buffer, and generates multiple possible next-state samples for each state-action pair:

$$\tilde{s}_{t+1}^{(1)}, \tilde{s}_{t+1}^{(2)}, \dots, \tilde{s}_{t+1}^{(K)} \sim \tilde{p}_0(\cdot\,|s_t, a_t) \tag{45}$$

Where $\widetilde{p_0}(\cdot\,|s_t, a_t)$ denotes the empirical nominal transition distribution approximated by the transition VAE. These generated next-state samples are used to approximate the expectation term in the critic target. It should be noted that the transition VAE does not serve as a load demand or PV generation forecasting model for actual scheduling, but is instead used for critic target estimation during the training process.

In addition, DR-SAC uses a G-network to approximate the dual variable required in the robust critic target, avoiding the need to repeatedly solve a worst-case optimization problem for each state-action pair in the continuous state-action space. The G-network takes the current state-action pair (s_t, a_t) as input and outputs the corresponding dual variable:

$$\beta_t = g_\eta(s_t, a_t) \tag{46}$$

where g_η denotes the G-network parameterized by η, and β_t is the dual variable corresponding to the current state-action pair. This dual variable corresponds to the KL-divergence constraint and is used to approximate the worst-case expectation computation. Through the G-network, DR-SAC improves the computational efficiency of robust critic updates in continuous state and action spaces.

In summary, the key modifications of DR-SAC are concentrated in the critic target. The transition VAE is responsible for generating possible next-state samples to approximate the expectation term; the G-network estimates the dual variable to help obtain a more conservative robust target; and the critic network is updated based on this worst-case expected future value. Through this design, DR-SAC is able to improve the robustness of the learned policy against transition uncertainty without modifying the original BESS scheduling environment, state definition, action space, or reward function.

3.4.3 Network Architecture

The DR-SAC agent in this study consists of four neural network components. The actor network takes the current state as input and outputs a continuous BESS charging or discharging action using a Gaussian policy. The twin critic networks take a state-action pair as input and estimate the corresponding Q-value, with the minimum of the two values taken to mitigate Q-value overestimation. The parameters of the target critic networks are updated using soft updates based on exponential moving average to stabilize training. The transition VAE learns the latent distribution of state transitions from the replay buffer, generating K next-state samples for each state-action pair to approximate the expectation term under the nominal transition distribution. The G-network takes a state-action pair as input and outputs an approximation of the dual variable β, with a softplus activation function [75] applied at the output layer to ensure $\beta > 0$. The update procedure for each component is detailed in Section 3.4.4.

3.4.4 Training Procedure

The DR-SAC training procedure in this study consists of three phases.

In Phase 1, offline data collection is conducted using a random policy. The agent interacts with the training environment and randomly generates different BESS charging and discharging actions, producing an offline dataset that covers diverse operating conditions. The collected transitions are then stored in the replay buffer. This design increases the diversity of state-action transitions in the dataset and provides the basis for subsequent offline DR-SAC training.

In Phase 2, the VAE is pre-trained to learn the latent distribution of state transitions from the replay buffer, providing reliable next-state samples for subsequent robust critic updates.

In Phase 3, offline DR-SAC training is conducted by sampling mini-batches from the replay buffer and sequentially updating the VAE, G-network, twin critic networks, and actor network. A portion of the training data is held out as a validation set. Early stopping is applied to retain the model checkpoint with the lowest validation cost as the final scheduling policy. Once offline training is complete, the model parameters are fixed and directly deployed for scheduling without further online updates.

The key hyperparameters are summarized in Table 3.4.

Table 3.4 Hyperparameter settings

Parameter	Value
Discount factor (γ)	0.99
Soft update rate (τ)	0.005

Entropy temperature (α)	0.05
Robust radius (δ)	0.01
VAE samples (K)	10
VAE latent dimension	16
Hidden dimension	256
Batch size	256
Replay buffer capacity	200000
Learning rate	3×10^{-4}
VAE pretraining steps	10000
Total training updates	300000

3.5 Experimental Setup and Evaluation Design

This section describes the experimental setup and evaluation design of this study. To ensure a consistent basis for comparison among different BESS scheduling methods, all methods are built upon the same microgrid system configuration, BESS technical specifications, TOU tariff, contract capacity rules, battery degradation model, and TREC revenue calculation. The differences among methods lie primarily in their scheduling decision mechanisms and whether they rely on future or forecast information.

This study uses one full year of real hourly data from the NTUST campus as the case study dataset, comprising building load demand, PV generation, and hourly electricity prices constructed based on the Taiwan TOU tariff. All data are recorded at an hourly resolution.

In terms of experimental design, the data are divided into a training period and a testing period. The training period is used for policy learning in SAC and DR-SAC, while the testing period is used for the final evaluation of all methods. The testing period comprises 50 days, and the training period consists of the remaining 315 days outside the testing period. All methods are evaluated under the same testing period to ensure consistency in cost comparison.

In MILP-based methods, the optimization model directly solves for the BESS charging and discharging schedule based on the objective function and system constraints defined in Section 3.2. In RL-based methods, the SAC and DR-SAC agents output BESS actions based on the current state, and the environment computes the reward and next state using the same system constraints, SOC transition, and cost calculation logic. Therefore, although MILP-based and RL-based methods differ in their decision-making mechanisms, their evaluation basis remains consistent.

The experimental design serves three main purposes. First, the theoretical optimum establishes the minimum achievable cost under complete future information, serving as a benchmark to measure the cost gap between SAC, DR-SAC, and other practical scheduling methods and the optimal solution. Second, SAC with online training and DR-SAC with offline robust training are compared under the same testing conditions in terms of cost performance and operational behavior, in order to evaluate whether the distributionally robust formulation improves the BESS scheduling policy. Third, forecast error modeling is used to compare the robustness of RL-based methods and forecast-based MILP scheduling methods under different levels of forecast error.

3.5.1 Comparison Methods

To evaluate the cost robustness of different scheduling methods under forecast errors, this study incorporates day-ahead scheduling and model predictive control (MPC) as MILP-based scheduling benchmarks. MPC has been widely discussed as a primary scheduling framework for handling forecast uncertainty in microgrid energy management [76], and MILP-based day-ahead scheduling has been broadly applied to BESS charging and discharging scheduling [77].

In day-ahead scheduling, the system solves for a complete 24-hour BESS charging and discharging schedule at the beginning of each day, using the forecasted building load demand and PV generation as inputs, subject to the objective function and system constraints defined in Section 3.2. The resulting schedule is executed hourly throughout the day against actual load and generation data. Since the schedule is derived from forecast data, deviations between forecasts and actual values lead to suboptimal execution outcomes and increased operating costs.

In MPC, the system adopts a receding horizon architecture, solving for a 24-hour BESS charging and discharging schedule at each hourly time step using the current forecast data, but only executing the decision of the first time step. At the next time step, updated forecast data are obtained and the optimization is solved again [78]. This receding horizon mechanism allows MPC to continuously incorporate the latest forecast information, providing greater forecast error correction capability compared to day-ahead scheduling.

Both methods adopt the same cost calculation framework as defined in Section 3.2, including electricity purchase cost, battery degradation cost, contract capacity cost, and TREC revenue, ensuring a consistent evaluation basis with the theoretical optimum, SAC, and DR-SAC.

3.5.2 Forecast Error Modeling

To evaluate the cost robustness of different scheduling methods under forecast errors, this study simulates forecast errors for building load demand and PV generation at varying error levels using normally distributed noise, following commonly adopted error simulation approaches in the literature[79]-[81]. Prior studies have indicated that forecast errors for load demand and PV generation can be treated as mutually independent random variables that approximately follow a normal distribution [69]. Accordingly, independent normally distributed noise is applied separately to simulate forecast errors for both variables.

Forecast errors for both load demand and PV generation are modeled using a relative error approach. For each actual data point, the forecast value is generated by multiplying the actual value by $(1+\varepsilon)$, where $\varepsilon \sim \mathcal{N}(0,(x/100)^2)$ and x denotes the corresponding error level parameter. This design ensures that forecast errors scale proportionally with actual values, reasonably reflecting forecast uncertainty across different load and generation levels, while ensuring that no spurious non-zero forecast values are generated during nighttime periods when PV generation is zero. All forecast values are subject to a non-negativity constraint to comply with the physical characteristics of power systems. The actual MAPE corresponding to each error level is computed ex post from the simulated data, where the PV forecast MAPE is calculated only over time periods with non-zero PV generation to exclude the influence of nighttime zero values on error statistics.

This study designs 20 error levels, with the error parameter x incrementing from 1 to 20, corresponding to load forecast MAPE values ranging from 1.01% to 19.95% and PV forecast MAPE values ranging from 1.12% to 22.87%, as shown in Table 3.5. Each error level produces a corresponding forecast dataset, which is provided separately to day-ahead scheduling and MPC as input to evaluate the total operating cost performance of each method under different forecast accuracy conditions.

Fig. 3.3 and Fig. 3.4 present the forecast data distributions for load demand and PV generation, respectively, across different error levels for a representative day. As the error level increases, the deviation between forecast and actual values progressively widens, with forecast curves exhibiting greater fluctuation at higher error levels. PV generation forecast values remain at zero during nighttime periods, confirming that the relative error modeling approach is consistent with the physical characteristics of power systems.

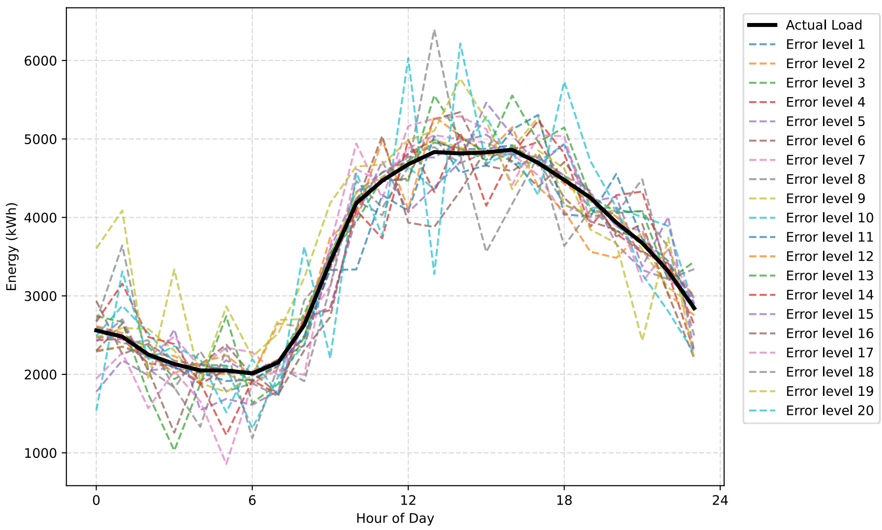

Fig. 3.3 Load Forecast Comparison under Different Error Level

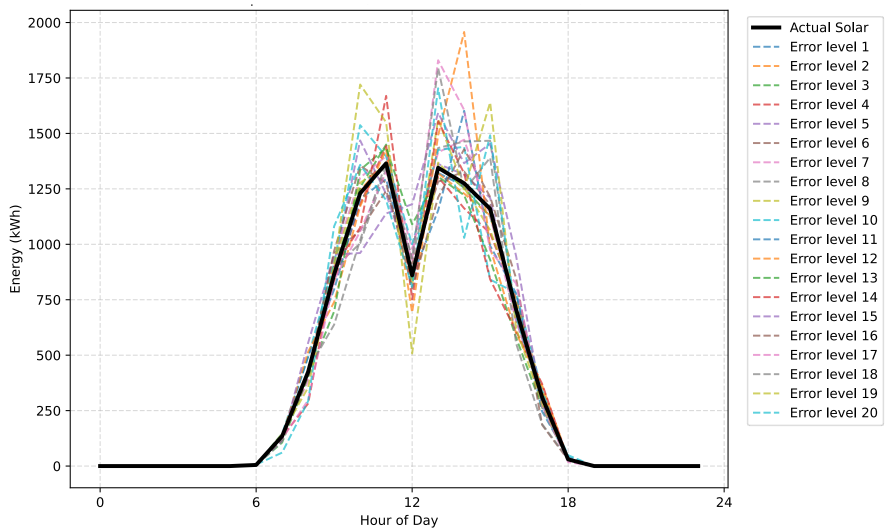

Fig. 3.4 PV Generation Forecast Comparison under Different Error Levels

Table 3.5 Forecast error levels and corresponding MAPE values

Error Level	Load Forecast MAPE	PV Forecast MAPE
1	1.01%	1.12%
2	2.02%	2.32%
3	2.93%	3.40%
4	3.98%	4.62%
5	5.04%	5.67%
6	6.08%	6.79%
7	7.15%	7.96%
8	8.16%	9.15%
9	8.95%	10.19%
10	10.06%	11.33%
11	11.20%	12.68%
12	12.01%	13.76%
13	13.04%	14.79%
14	13.81%	15.69%
15	14.61%	17.03%
16	16.28%	17.77%

17	17.17%	19.28%
18	17.94%	20.26%
19	19.21%	21.42%
20	19.95%	22.87%

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Case Study

This study uses the NTUST campus dataset as the case study. The dataset contains one year of hourly data, including building load demand, PV generation, and TOU electricity prices. These data serve as the main inputs for simulating the battery scheduling methods.

Fig. 4.1 presents the annual profiles of the dataset, including both hourly values and daily average trends. As shown in the figure, the load demand exhibits clear seasonal variations. The overall electricity demand is higher during the summer period, while it is relatively lower during the non-summer period. The daily average curve indicates that electricity demand varies with seasonal conditions and operational patterns, reflecting the load fluctuation characteristics of a campus energy system in real-world operation.

Compared with load demand, PV generation shows more significant intermittency and variability, mainly due to solar irradiation conditions and weather changes. PV generation is generally higher and more stable during the summer period, whereas consecutive days with particularly low PV generation are more frequently observed in winter. Although PV generation can provide renewable energy for the microgrid, its uncertain output characteristics also increase the complexity of BESS scheduling.

The electricity price follows a TOU pricing structure, varying across seasons and time periods. In particular, the summer peak period has a higher electricity price, which provides an economic incentive for BESS scheduling. Ideally, the BESS can be charged during low-price periods and discharged during high-price periods to reduce electricity purchasing costs.

The technical specifications of the BESS are listed in Table 4.1. Python was used as the main experimental environment in this study. The mathematical programming models were solved using the Gurobi Optimizer[82].

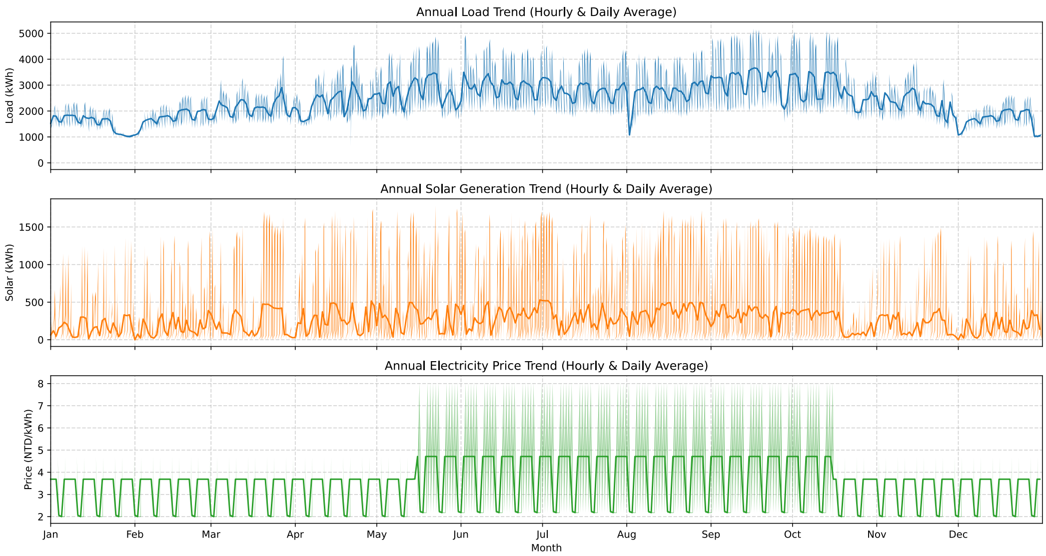

Fig. 4.1 Annual profiles of the case study dataset

Table 4.1 Technical specification of the BESS

BESS specification	
Rated capacity	1000kWh
Rated power	750kW
Charge efficiency	0.95
Discharge efficiency	0.9
Minimum state of charge	0.2
Maximum state of charge	0.9

4.2 Cost Performance Evaluation

Fig. 4.2 shows the total operating cost over the 50-day testing period, including energy purchase cost, contract capacity cost, battery degradation cost, and TREC revenue. The results show that the total operating cost of SAC is 14.37 million NTD, which is approximately 5.46% higher than the theoretical optimum. In contrast, the total operating cost of DR-SAC is 13.925 million NTD, which is approximately 2.20% higher than the theoretical optimum. Compared with SAC, DR-SAC achieves a total cost closer to the theoretical minimum, indicating better cost control capability.

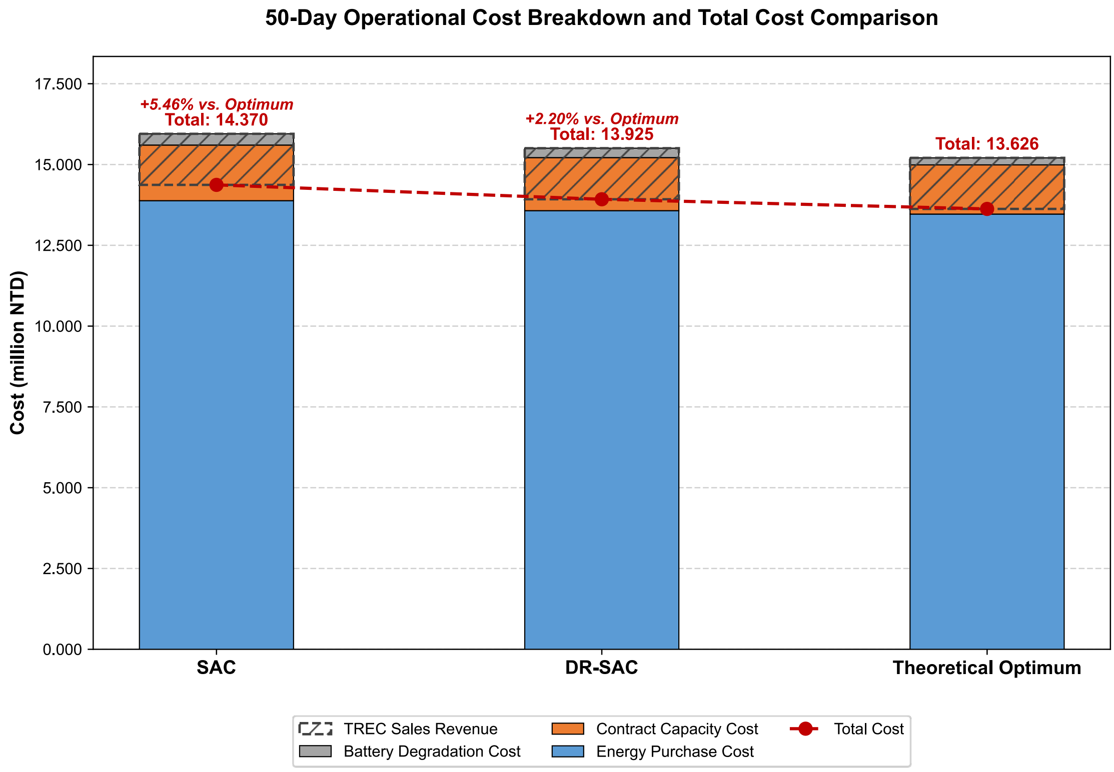

Fig. 4.2 Total cost breakdown and comparison

Table 4.2 presents the detailed cost breakdown of the three methods. For all methods, the major cost component is the energy purchase cost, followed by the contract capacity cost. SAC results in higher energy purchase and contract capacity costs than DR-SAC, suggesting that SAC is less effective in determining battery charging and discharging timing and in controlling peak grid demand. In contrast, DR-SAC effectively reduces both energy purchase cost and contract capacity cost, resulting in an overall performance closer to the theoretical optimum. In particular, the contract capacity cost is significantly lower for DR-SAC than for SAC, indicating that DR-SAC has better capability in suppressing peak grid power demand.

For battery degradation cost, the theoretical optimum has the lowest value, at 0.217 million NTD. DR-SAC results in a battery degradation cost of 0.292 million NTD, while SAC has the highest value, at 0.349 million NTD. This suggests that SAC may lead to more frequent or more aggressive charging and discharging behavior during scheduling, resulting in higher battery degradation cost. By contrast, DR-SAC reduces unnecessary battery cycling while lowering the overall operating cost, indicating better battery operation efficiency.

Table 4.2 Total operating cost (million NTD)

	SAC	DR-SAC	Theoretical Optimum
Energy purchase cost	13.881	13.572	13.466
Contract capacity cost	1.721	1.642	1.523
Battery degradation cost	0.349	0.292	0.217
TREC revenue	1.580	1.580	1.580
Total cost	14.370	13.925	13.626

The TREC sales revenue is the same for all three methods, at 1.580 million NTD. This is because TREC revenue mainly depends on the total PV generation during the testing period rather than the scheduling decisions of different methods. Therefore, under the setting of this study, TREC revenue does not directly contribute to the cost differences among the methods. Instead, it is treated as an identical revenue term deducted from the total operating cost.

The above results show that DR-SAC provides better cost control than SAC. The improvement does not come from a single cost component, but is reflected simultaneously in energy purchase cost, contract capacity cost, and battery degradation cost. The reduction in energy purchase cost indicates that DR-SAC can utilize the battery more effectively for time-shifting, thereby reducing electricity purchases during high-price periods. The reduction in contract capacity cost suggests that DR-SAC is more effective in controlling peak grid power demand. In addition, the lower battery degradation cost indicates that the charging and discharging behavior of DR-SAC is less aggressive, allowing it to reduce operating cost while avoiding unnecessary battery usage.

Overall, the cost performance evaluation demonstrates that DR-SAC provides a more economical scheduling strategy than SAC in this case study and achieves a total cost closer to the theoretical optimum. However, the cost results alone only indicate the overall economic performance of different methods and are not sufficient to explain the differences in their operational behavior. Therefore, the following section further analyzes the operational characteristics of different methods, including battery charging and discharging behavior, SOC variation, and contract capacity control, to explain the underlying reasons behind the cost differences.

4.3 Operational Characteristics

This section further analyzes the operational characteristics of different scheduling methods to explain the reasons behind the cost differences discussed in Section 4.2. In addition to the total operating cost, the actual charging and discharging behavior of the BESS, SOC variation, peak grid demand control, and battery cycling intensity all affect the energy purchase cost, contract capacity cost, and battery degradation cost. Therefore, this section analyzes the representative weekly scheduling behavior of DR-SAC, the seasonal cost reduction effect, contract capacity exceedance conditions, and battery cycling intensity.

Fig. 4.3 shows the scheduling behavior of DR-SAC during a representative summer week. As shown in the figure, DR-SAC adjusts the charging and discharging decisions of the BESS according to changes in electricity price, load demand, and PV generation. During periods with low electricity prices or high PV generation, the BESS tends to charge, causing the SOC to increase to a higher level. In contrast, during periods with high electricity prices or high load demand, the BESS discharges to reduce grid purchases. This operating pattern indicates that DR-SAC can utilize the BESS for energy time-shifting by transferring part of the low-cost electricity or renewable energy to periods with higher demand or higher electricity prices.

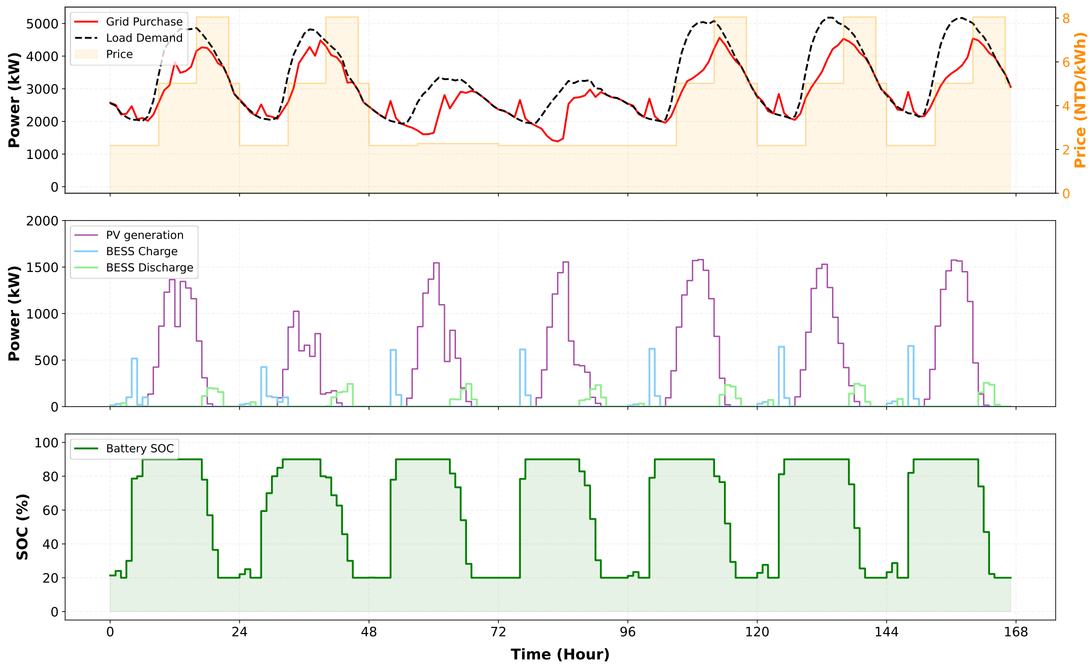

Fig. 4.3 DR-SAC scheduling behavior during the summer week

From the perspective of SOC variation, the SOC during the representative summer week shows a regular intraday cycling pattern. The BESS generally charges during low-price periods or periods with higher PV generation, and discharges during evening peak periods or high-price periods. This result indicates that DR-SAC does not simply perform frequent charging and discharging, but instead selects economically beneficial charging and discharging timing based on system conditions. Since load demand is higher in summer and the peak electricity price is more significant, the BESS has a stronger economic incentive for scheduling during the summer period.

Fig. 4.4 shows the scheduling behavior of DR-SAC during a representative non-summer week. Compared with summer, both load demand and electricity price levels are lower in the non-summer period. As a result, the variation in grid purchases is smaller, and the operation of the BESS is relatively less regular. As shown in the figure, DR-SAC still charges during some low-price periods or periods with higher PV generation, and discharges when the load demand is higher. However, the charging and discharging behavior in the non-summer period is more affected by PV generation fluctuations and load variations. The SOC variation also does not show the same highly regular intraday cycling pattern as in summer. This result reflects that the scheduling strategy of DR-SAC adapts to seasonal conditions. It performs more active time-shifting and peak control in summer, while adopting a more flexible operating behavior in the non-summer period.

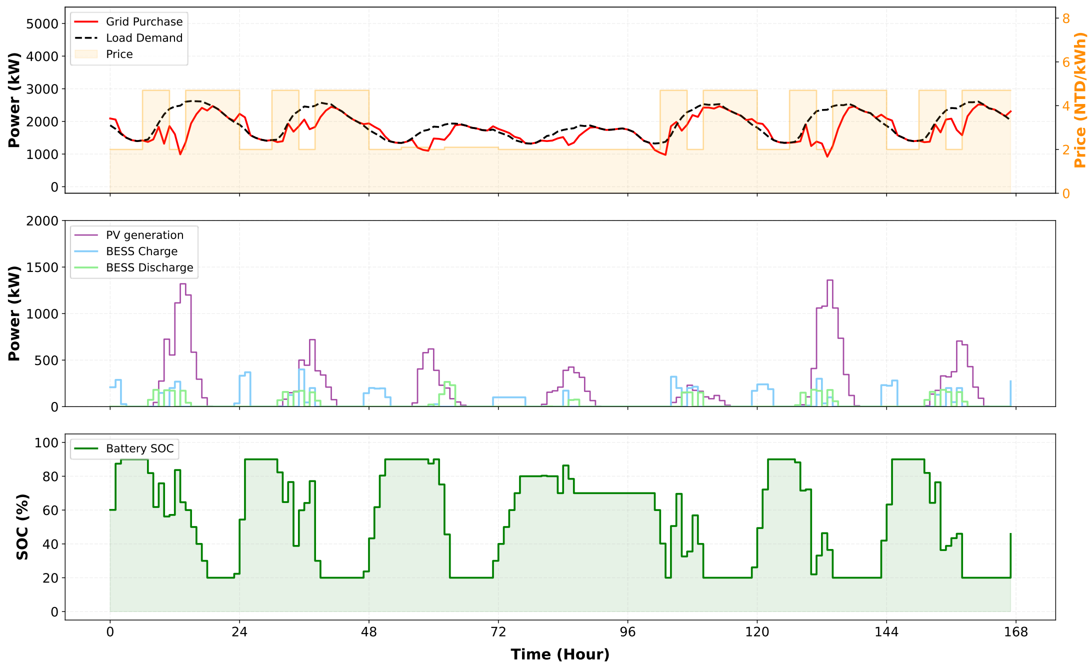

Fig. 4.4 DR-SAC scheduling behavior during the non-summer week

Table 4.3 presents the average weekly electricity cost of different methods in summer and non-summer periods, as well as the cost reduction rate compared with the scenario without BESS. In the summer scenario, the average weekly electricity cost without BESS is 2.511 million NTD. SAC reduces the average weekly electricity cost to 2.214 million NTD, corresponding to a reduction of 11.8%. DR-SAC further reduces the cost to 2.123 million NTD, with a reduction of 15.5%. The theoretical optimum achieves the lowest cost of 2.099 million NTD, corresponding to a reduction of 16.4%. These results show that under high-load and high-price summer conditions, DR-SAC achieves a significantly greater cost reduction than SAC and performs close to the theoretical optimum.

In the non-summer scenario, the average weekly electricity cost without BESS is 1.348 million NTD. SAC, DR-SAC, and the theoretical optimum reduce the average weekly electricity cost to 1.246 million NTD, 1.197 million NTD, and 1.189 million NTD, respectively. The corresponding cost reduction rates are 7.5%, 11.2%, and 11.7%. Compared with summer, both the average weekly electricity cost and the cost reduction magnitude are lower in the non-summer period. This is mainly because the non-summer period has lower load demand and less pressure from peak electricity prices, which limits the economic benefits that can be created by the BESS. Nevertheless, DR-SAC still maintains a performance close to the theoretical optimum, indicating that it provides a stable cost reduction effect across different seasonal conditions.

Table 4.3 Average weekly electricity charge and cost reduction by season

Method	Summer charge (million NTD)	Summer reduction	Non-summer charge (million NTD)	Non-summer reduction
No BESS	2.511	-	1.348	-
SAC	2.214	11.8%	1.246	7.5%
DR-SAC	2.123	15.5%	1.197	11.2%
Theoretical optimum	2.099	16.4%	1.189	11.7%

Fig. 4.5 further presents the average daily electricity cost by day of week in summer and non-summer periods. In summer, the average electricity cost on weekdays is significantly higher than that on weekends, reflecting higher campus load demand during weekdays. In addition, due to the higher peak electricity price in summer, the cost difference between weekdays and weekends is more pronounced. In the summer case, the scenario without BESS has the highest average electricity cost, while SAC, DR-SAC, and the theoretical optimum all reduce the average cost. Among them, DR-SAC performs close to the theoretical optimum on most weekdays, indicating better scheduling capability under high-load and high-price conditions.

In contrast, the average daily electricity cost in the non-summer period is generally lower than that in summer, and the difference between weekdays and weekends is smaller. However, the average cost on weekends is still clearly lower than that on weekdays, indicating that the campus load pattern is still affected by operating and non-operating days. DR-SAC still outperforms SAC on most days in the non-summer period and remains close to the theoretical optimum. This suggests that the improvement achieved by DR-SAC is not limited to summer peak conditions, but can also provide relatively stable cost reduction under non-summer conditions.

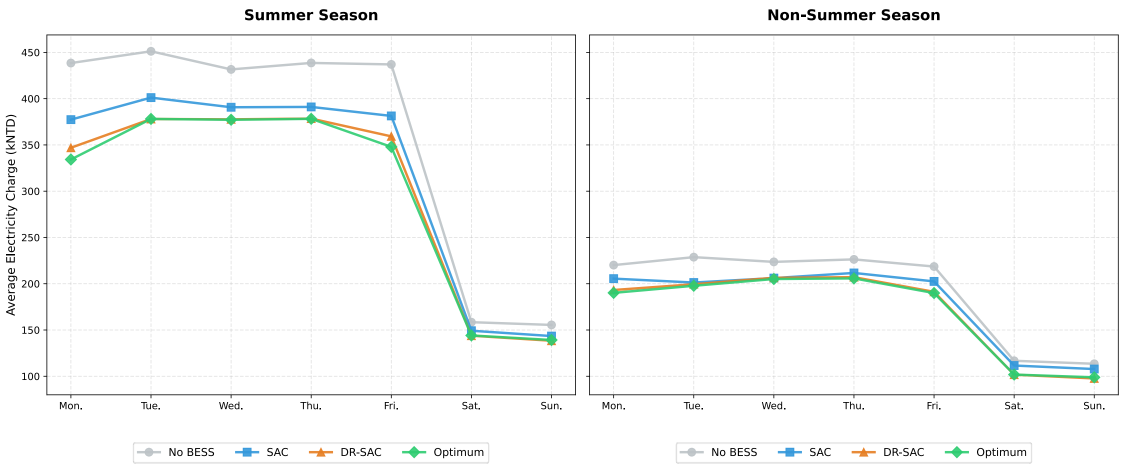

Fig. 4.5 Average electricity charge by day of week

Table 4.4 and Fig. 4.6 compare the performance of different methods in contract capacity control. In the scenario without BESS, contract capacity exceedance occurs 42 times during the 50-day testing period, and the maximum grid purchase reaches 5033 kW. This indicates that the system tends to exceed the contract capacity limit during high-load periods when battery scheduling is not available. SAC reduces the maximum grid purchase to 4842 kW, with 40 exceedance occurrences, showing only limited improvement. In contrast, DR-SAC reduces the exceedance count to 23 and lowers the maximum grid purchase to 4665 kW, indicating that it can more effectively suppress peak grid demand. The theoretical optimum does not have any contract capacity exceedance, and its maximum grid purchase is controlled at 4400 kW, reflecting the best peak control performance achieved by perfect-foresight optimization.

Fig. 4.6 shows that both the scenario without BESS and SAC exceed the contract capacity line multiple times, whereas DR-SAC has considerably fewer exceedance events and a relatively smoother grid purchase profile. This observation is consistent with the exceedance counts reported in Table 4.4 and indicates that the lower contract capacity cost of DR-SAC mainly comes from better peak control. However, DR-SAC still cannot completely avoid exceedance, suggesting that there remains a gap between DR-SAC and the theoretical optimum in precise peak grid demand control.

Table 4.4 Contract capacity exceedance summary

Scenario	Number of exceedances	Maximum grid purchase (kW)

No BESS	42	5033
SAC	40	4842
DR-SAC	23	4665
Theoretical optimum	0	4400

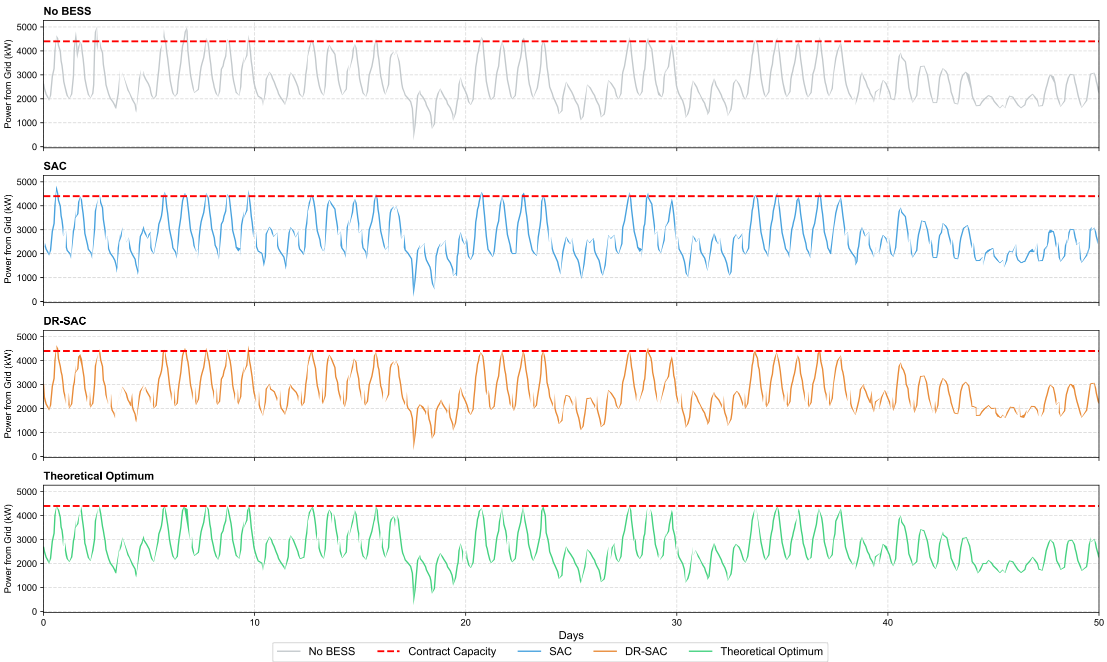

Fig. 4.6 Hourly grid purchase and contract capacity

Table 4.5 presents the battery equivalent full cycles of different methods over the 50-day testing period. The equivalent full cycles of SAC are 74.00, while those of DR-SAC and the theoretical optimum are 61.80 and 45.92, respectively. This result indicates that SAC uses the battery most frequently, which may lead to a higher battery degradation cost. The equivalent full cycles of DR-SAC are lower than those of SAC, suggesting that DR-SAC can reduce unnecessary battery cycling while lowering the operating cost. The theoretical optimum has the lowest equivalent full cycles, indicating that the optimization model can utilize the battery more efficiently under perfect information.

Overall, the above results help explain the better cost performance of DR-

SAC reported in Section 4.2. Compared with SAC, DR-SAC not only reduces energy purchase cost, but also decreases the number of contract capacity exceedances and the degree of battery cycling. This suggests that the improvement of DR-SAC does not simply come from using the BESS more frequently. Instead, it results from more effective charging and discharging timing, allowing DR-SAC to achieve a better balance among cost reduction, peak control, and battery usage. However, there is still a gap between DR-SAC and the theoretical optimum, especially in completely avoiding contract capacity exceedance and further reducing battery cycling. This also indicates that although DR-SAC can develop an effective operating strategy without complete future information, it still cannot fully reach the operational performance of perfect-foresight optimization.

Table 4.5 Battery equivalent full cycles summary

Method	Equivalent full cycles
SAC	74.00
DR-SAC	61.80
Theoretical optimum	45.92

4.4 Comparison with MILP-Based Scheduling

The previous results have shown that DR-SAC provides better cost control over the 50-day testing period. This section further investigates how the cost performance of day-ahead scheduling and MPC changes when forecast errors exist in load demand and PV generation, and compares them with SAC, DR-SAC, and the theoretical optimum. The load forecast MAPE and PV forecast MAPE corresponding to each forecast error level are presented in table 3.5, which serves as a reference for interpreting the forecast error levels in this section.

Fig. 4.7 shows the total cost of different methods over the 50-day testing period under different forecast error levels. The theoretical optimum represents the optimization result under complete future information; therefore, it is not affected by forecast errors and maintains the lowest cost. SAC and DR-SAC do not use future forecast data as decision inputs, so their total costs remain unchanged across different forecast error levels. In contrast, the total costs of day-ahead scheduling and MPC both increase as the forecast error level increases, with day-ahead scheduling showing a more pronounced increase than MPC.

As shown in the figure, MPC can still maintain a cost performance lower than or close to DR-SAC when the forecast error level is between levels 1 and 5. This range corresponds to load forecast MAPE values of 1.01%–5.04% and PV forecast MAPE values of 1.12%–5.67%. However, when the forecast error level increases to level 6, corresponding to a load forecast MAPE of 6.08% and a PV forecast MAPE of 6.79%, the total cost of MPC starts to exceed that of DR-SAC. This result indicates that the cost performance of DR-SAC is already close to MPC under low forecast errors. Once the forecast error slightly increases, DR-SAC begins to show a cost advantage over MPC.

Day-ahead scheduling loses its cost advantage over DR-SAC even earlier. When the forecast error level reaches level 3, corresponding to a load forecast MAPE of 2.93% and a PV forecast MAPE of 3.40%, the total cost of day-ahead scheduling already becomes higher than that of DR-SAC. This indicates that day-ahead scheduling is more sensitive to forecast errors than MPC.

Compared with DR-SAC, SAC shows weaker cost performance and therefore only outperforms MILP-based scheduling under higher forecast error levels. The total cost of day-ahead scheduling becomes higher than SAC after approximately forecast error level 13, corresponding to a load forecast MAPE of 13.04% and a PV forecast MAPE of 14.79%. MPC becomes higher than SAC after approximately forecast error level 14, corresponding to a load forecast MAPE of 13.81% and a PV forecast MAPE of 15.69%. In other words, although SAC is not affected by forecast errors, its cost advantage mainly appears when forecast errors are relatively high. In contrast, DR-SAC can outperform MILP-based scheduling at a much lower forecast error threshold.

Overall, Fig. 4.7 shows that DR-SAC achieves cost performance close to, and even better than, MPC under relatively accurate forecasts, whereas SAC only shows a relative advantage after forecast errors further increase. This indicates that DR-SAC narrows the cost gap with MILP-based scheduling more effectively than SAC and maintains competitive scheduling performance without relying on future forecast data.

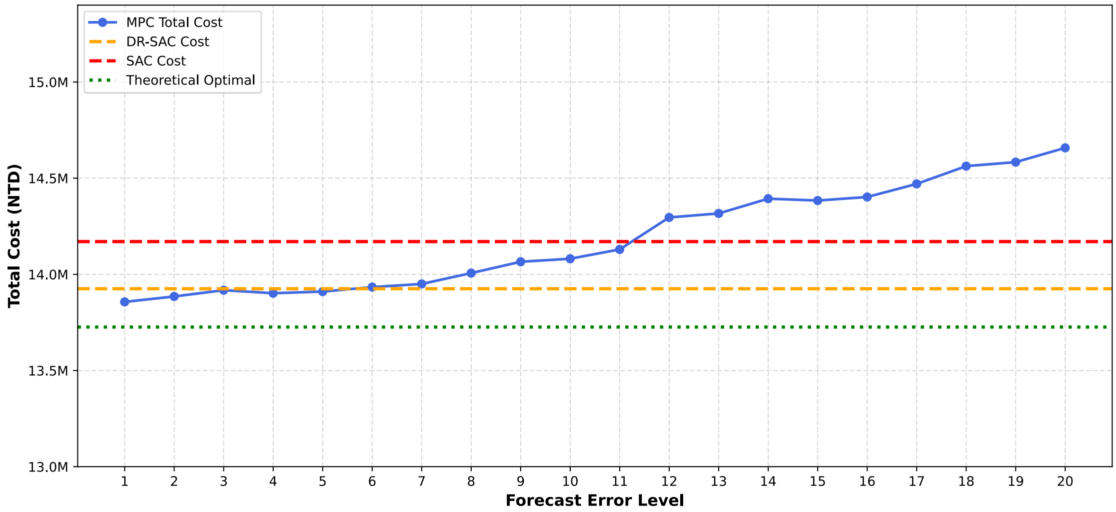

Fig. 4.7 Impact of forecast error on mathematical programming-based scheduling

CHAPTER 5
CONCLUSION

5.1 Research Summary

This study focuses on a grid-connected PV-BESS microgrid and investigates the battery charging and discharging scheduling problem under uncertain operating conditions. Since PV generation and building load demand are time-varying and subject to forecast uncertainty, BESS scheduling should not only reduce electricity purchase costs, but also consider contract capacity-related costs, battery degradation costs, and renewable energy certificate revenue. Therefore, this study establishes a total operating cost evaluation framework that integrates electricity purchase cost, contract capacity cost, battery degradation cost, and TREC revenue. The NTUST campus dataset is used as the case study to analyze PV-BESS microgrid scheduling.

In terms of methodology, this study first constructs a MILP-based theoretical optimum as an idealized solution under complete future information, which is used to evaluate the cost gap between RL-based scheduling methods and the theoretical optimum. SAC and DR-SAC are then applied to the BESS scheduling problem, allowing the agent to make continuous charging and discharging decisions based on the current system state without relying on complete future information. DR-SAC further incorporates transition distribution uncertainty into the critic update process, enabling the model to evaluate future values in a more conservative and stable manner when facing variations in load demand, PV generation, and system states. This improves the robustness of the scheduling policy under uncertain operating environments.

5.2 Main Findings

The results show that DR-SAC achieves better total cost performance than standard SAC, and its result is closer to the MILP-based theoretical optimum. During the 50-day testing period, the total operating costs of SAC, DR-SAC, and the theoretical optimum are 14.370, 13.925, and 13.626 million NTD, respectively. This indicates that DR-SAC can effectively reduce BESS scheduling costs without relying on complete future information, while achieving a better balance among electricity purchase cost, contract capacity cost, and battery degradation cost.

In terms of contract capacity control, DR-SAC effectively reduces the maximum grid purchase power of the system. Without BESS, the maximum grid purchase power reaches 5033 kW. SAC reduces it to 4842 kW, while DR-SAC further reduces it to 4665 kW, showing that DR-SAC provides better peak demand suppression. Compared with the no BESS case, DR-SAC saves 0.388 million NTD per week on average in the summer case and 0.151 million NTD per week on average in the non-summer case. This demonstrates that BESS scheduling can provide electricity cost reduction benefits under different seasonal conditions.

Furthermore, the forecast error analysis shows that when forecast errors increase, the cost performance of mathematical programming-based scheduling is significantly affected. This reflects that such methods, in practical scheduling applications, depend to some extent on future input information and forecast quality. In contrast, SAC and DR-SAC do not directly rely on future forecast data as decision inputs, and therefore can maintain more stable cost performance under different forecast error levels. Overall, compared with SAC, DR-SAC provides better cost control, peak demand suppression, and operational robustness, although a certain performance gap remains compared with perfect-foresight optimization.

5.3 Limitation and Future Work

This study still has several limitations. First, the simulation environment was constructed using historical data, and no actual field deployment was conducted. Therefore, practical factors such as physical device control, communication latency, measurement errors, and hardware integration were not considered. Second, this study used the campus dataset as the case study. As a result, the findings mainly reflect the scheduling performance of a campus-based PV-BESS microgrid under a given system configuration and tariff structure. If the proposed method is applied to residential, commercial, or industrial users, or to cases with different electricity tariff structures, further validation is still required. In addition, in the forecast error analysis, this study mainly controlled the forecast errors of PV generation and load demand at similar levels in order to observe the cost variation of mathematical programming-based scheduling under increasing overall forecast uncertainty. However, the individual effects of these two types of forecast errors were not further separated. Therefore, the sensitivity of electricity purchase cost, peak demand control, and BESS operation behavior to different sources of uncertainty remains to be further investigated. On the other hand, the model in this study focuses on BESS energy scheduling and total operating cost minimization. Therefore, power system-level issues such as distribution network power flow, voltage stability, frequency control, and power quality were not included. Although calendar degradation and cycle degradation were incorporated into the cost evaluation, the battery degradation model is still a cost-based simplified representation and does not consider the nonlinear effects of temperature, depth of discharge, and state-of-health variation on battery degradation.

Future research can be extended in several directions. First, this study adopted an hourly time step for scheduling. If the time resolution is further increased to the minute level or an even finer scale, short-term fluctuations in PV generation and load demand will become more evident, and the uncertainty caused by forecast errors may also become more significant. Under such conditions, scheduling methods should not only reduce immediate operating costs, but also preserve sufficient flexibility for subsequent operations. Therefore, distributionally robust scheduling designs may have greater practical value. Second, as the number of devices in a microgrid increases and the scheduling time step becomes shorter, the number of decision variables, the scale of constraints, and the computational burden of mathematical programming methods may increase accordingly. In comparison, once a RL method has been trained, it can quickly generate control decisions based on the current system state. Therefore, RL has potential for application in microgrid scheduling problems with high time resolution and high system complexity. However, its training cost, policy stability, and cross-scenario generalization capability still require further validation.

Furthermore, future studies can apply the proposed scheduling method to an actual microgrid field site or a hardware-in-the-loop testing environment to verify the feasibility and stability of DR-SAC in real systems. Future research may also further separate the individual effects of PV forecast error and load forecast error, and analyze the sensitivity of scheduling cost, peak demand control, and BESS operation behavior to different sources of uncertainty. In terms of the battery model, a more detailed battery aging model can be introduced in future work by incorporating the nonlinear effects of temperature, charge-discharge rate, and depth of discharge into the scheduling objective, to evaluate the long-term economic benefits of different scheduling strategies more accurately. Finally, the proposed framework can also be applied to different types of microgrid scenarios, including residential, commercial, and industrial cases, or cases with different electricity tariff structures, to evaluate the cross-scenario applicability of DR-SAC.

REFERENCE

[1] Eyimaya, S. E., & Altin, N. (2024). Review of energy management systems in microgrids. *Applied Sciences, 14*(3), 1249. https://doi.org/10.3390/app14031249

[2] Ali, M. A., Besheer, A. H., Emara, H. M., & Bahgat, A. (2025). Optimal micro-grid battery scheduling within a comprehensive smart pricing scheme. *Scientific Reports, 15*, Article 20098. https://doi.org/10.1038/s41598-025-02690-9

[3] Zhao, C., & Li, X. (2024). Microgrid optimal energy scheduling considering neural network based battery degradation. IEEE Transactions on Power Systems, 39(1), 1594–1607. https://doi.org/10.1109/TPWRS.2023.3239113

[4] Zia, M. F., Elbouchikhi, E., & Benbouzid, M. (2018). Microgrids energy management systems: A critical review on methods, solutions, and prospects. Applied Energy, 222, 1033–1055. https://doi.org/10.1016/j.apenergy.2018.04.103

[5] Liu, D., Zang, C., Zeng, P., Li, W., Wang, X., Liu, Y., & Xu, S. (2023). Deep reinforcement learning for real-time economic energy management of microgrid system considering uncertainties. Frontiers in Energy Research, 11, Article 1163053. https://doi.org/10.3389/fenrg.2023.1163053

[6] Wang, Z., Cao, R., Tang, D., Wang, C., Liu, X., & Hu, W. (2025). Distributionally robust energy optimization with renewable resource uncertainty. Mathematics, 13(6), Article 992. https://doi.org/10.3390/math13060992

[7] Tan, B., Lin, Z., Zheng, X., & Xiao, F. (2023). Distributionally robust energy management for multi-microgrids with grid-interactive EVs considering the multi-period coupling effect of user behaviors. Applied Energy, 350, Article 121770. https://doi.org/10.1016/j.apenergy.2023.121770

[8] Cui, M., Zhou, D., Han, Y., Hanasusanto, G. A., Wang, Q., Zhang, H., & Zhou, Z. (2026). DR-SAC: Distributionally robust soft actor-critic for reinforcement learning under uncertainty. In International Conference on Learning Representations (ICLR 2026). https://openreview.net/forum?id=a19MA0ksbc

[9] National Renewable Energy Certification Center. Implementation regulations

governing renewable energy certificates. https://www.trec.org.tw/en/regulations/en-implementation-regulations-governing-renewable-energy-certificates

[10] Mazidi, M., Khezri, R., Mohiti Ardakani, M., Le, A. T., & Steen, D. (2023). Effects of calendar and cycle ageing on battery scheduling for optimal energy management: A case study of HSB Living Lab. In 2023 IEEE International Conference on Energy Technologies for Future Grids (ETFG). https://doi.org/10.1109/ETFG55873.2023.10407383

[11] Schonenberg, J., Mathur, K., Ros Perez, C., Hohl, D., Plaat, A., Moerland, T., & Michler, C. (2024). Benchmarking deep reinforcement learning for battery dispatch optimisation. In Proceedings of BNAIC/BeNeLearn 2024. https://bnaic2024.sites.uu.nl/wp-content/uploads/sites/986/2024/10/Benchmarking-Deep-Reinforcement-Learning-for-Battery-Dispatch-Optimisation.pdf

[12] Langer, L., & Volling, T. (2022). A reinforcement learning approach to home energy management for modulating heat pumps and photovoltaic systems. Applied Energy, 327, 120020. https://doi.org/10.1016/j.apenergy.2022.120020

[13] Olivares, D. E., Mehrizi-Sani, A., Etemadi, A. H., Cañizares, C. A., Iravani, R., Kazerani, M., Hajimiragha, A. H., Gomis-Bellmunt, O., Saeedifard, M., Palma-Behnke, R., Jiménez-Estévez, G. A., & Hatziargyriou, N. D. (2014). Trends in microgrid control. IEEE Transactions on Smart Grid, 5(4), 1905–1919. https://doi.org/10.1109/TSG.2013.2295514

[14] Das, C. K., Bass, O., Kothapalli, G., Mahmoud, T. S., & Habibi, D. (2018). Overview of energy storage systems in distribution networks: Placement, sizing, operation, and power quality. *Renewable and Sustainable Energy Reviews, 91*, 1205–1230.

[15] Azuatalam, D., Paridari, K., Ma, Y., Förstl, M., Chapman, A. C., & Verbič, G. (2019). Energy management of small-scale PV-battery systems: A systematic review considering practical implementation, computational requirements, quality of input data and battery degradation. *Renewable and Sustainable Energy Reviews, 112*, 555–570. https://doi.org/10.1016/j.rser.2019.06.007

[16] Husein, M., & Chung, I.-Y. (2018). Optimal design and financial feasibility of a

university campus microgrid considering renewable energy incentives. *Applied Energy*, 225, 273–289. https://doi.org/10.1016/j.apenergy.2018.05.036

[17] Lin, W.-R. (2025). *Design and evaluation of a smart energy management microgrid system for increasing renewable energy share in nearly-zero campus buildings* [Master's thesis, National Taiwan University of Science and Technology].

[18] Yang, H.-Y. (2024). *Reducing electricity bills through optimal contract power and energy storage system sizing and battery scheduling* [Master's thesis, National Taiwan University of Science and Technology].

[19] Taiwan Power Company. Electricity price list. Retrieved from https://www.taipower.com.tw/2289/2290/46940/

[20] Antoniadou-Plytaria, K., Steen, D., Le, A. T., Carlson, O., & Fotouhi Ghazvini, M. A. (2021). Market-based energy management model of a building microgrid considering battery degradation. IEEE Transactions on Smart Grid, 12(2), 1794–1804. https://doi.org/10.1109/TSG.2020.3037120

[21] Ju, C., Wang, P., Goel, L., & Xu, Y. (2018). A two-layer energy management system for microgrids with hybrid energy storage considering degradation costs. IEEE Transactions on Smart Grid, 9(6), 6047–6057. https://doi.org/10.1109/TSG.2017.2703126

[22] Lin, T.-L., Chen, K.-T., & Chen, Y.-H. (2025). Renewable energy certificate mechanism and markets in Taiwan: The evolution and characteristics. Energy Policy, 205, Article 114689. https://doi.org/10.1016/j.enpol.2025.114689

[23] National Renewable Energy Certification Center. (n.d.). National Renewable Energy Certification Center｜T-REC. https://www.trec.org.tw/en

[24] Azuatalam, D., Paridari, K., Ma, Y., Förstl, M., Chapman, A. C., & Verbič, G. (2019). Energy management of small-scale PV-battery systems: A systematic review considering practical implementation, computational requirements, quality of input data and battery degradation. Renewable and Sustainable Energy Reviews, 112, 555–570. https://doi.org/10.1016/j.rser.2019.06.007

[25] Zhao, C., & Li, X. (2024). Microgrid optimal energy scheduling considering neural network based battery degradation. IEEE Transactions on Power Systems, 39(1), 1594–1606. https://doi.org/10.1109/TPWRS.2023.3239113

[26] Zhang, Y., Wang, R., Zhang, T., Liu, Y., & Guo, B. (2016). Model predictive control-based operation management for a residential microgrid with considering forecast uncertainties and demand response strategies. IET Generation, Transmission & Distribution, 10(10), 2367–2378. https://doi.org/10.1049/iet-gtd.2015.1127

[27] Kou, P., Liang, D., & Gao, L. (2018). Stochastic energy scheduling in microgrids considering the uncertainties in both supply and demand. IEEE Systems Journal, 12(3), 2589–2600. https://doi.org/10.1109/JSYST.2016.2614421

[28] Hu, J., Shan, Y., Yang, Y., Parisio, A., Li, Y., Amjady, N., Islam, S. M., Cheng, K. W. E., Guerrero, J. M., & Rodríguez, J. (2024). Economic model predictive control for microgrid optimization: A review. IEEE Transactions on Smart Grid, 15(1), 472–484. https://doi.org/10.1109/TSG.2023.3266253

[29] Olivares, D. E., Lara, J. D., Cañizares, C. A., & Kazerani, M. (2015). Stochastic-predictive energy management system for isolated microgrids. *IEEE Transactions on Smart Grid, 6*(6), 2681–2693. https://doi.org/10.1109/TSG.2015.2469631

[30] Choi, J., Lee, J. I., Lee, I. W., & Cha, S. W. (2022). Robust PV-BESS scheduling for a grid with incentive for forecast accuracy. *IEEE Transactions on Sustainable Energy, 13*(1), 567–578. https://doi.org/10.1109/TSTE.2021.3120451

[31] Lyu, C., Jia, Y., & Xu, Z. (2021). Real-time operation optimization of microgrids with battery energy storage system: A tube-based model predictive control approach. *IEEE Transactions on Smart Grid, 12*(5), 4404–4417. https://doi.org/10.1109/TSG.2021.3086911

[32] Masa-Bote, D., Castillo-Cagigal, M., Matallanas, E., Caamaño-Martín, E., Gutiérrez, A., Monasterio-Huelin, F., & Jiménez-Leube, J. (2014). Improving photovoltaics grid integration through short time forecasting and self-consumption. *Applied Energy, 125*, 103–113. https://doi.org/10.1016/j.apenergy.2014.03.045

[33] Morstyn, T., Hredzak, B., Aguilera, R. P., & Agelidis, V. G. (2018). Model predictive control for distributed microgrid battery energy storage systems. IEEE Transactions on Control Systems Technology, 26(3), 1107–1114. https://doi.org/10.1109/TCST.2017.2699159

[34] Li, B., Chen, T., Wang, X., & Giannakis, G. B. (2019). Real-time energy management in microgrids with reduced battery capacity requirements. *IEEE Transactions on Smart Grid, 10*(2), 1928–1938. https://doi.org/10.1109/TSG.2017.2771399

[35] Kuznetsova, E., Li, Y.-F., Ruiz, C., Zio, E., Ault, G., & Bell, K. (2013). Reinforcement learning for microgrid energy management. *Energy, 59*, 133–146. https://doi.org/10.1016/j.energy.2013.05.060

[36] Nakabi, T. A., & Toivanen, P. (2021). Deep reinforcement learning for energy management in a microgrid with flexible demand. Sustainable Energy, Grids and Networks, 25, Article 100413. https://doi.org/10.1016/j.segan.2020.100413

[37] Kang, H., Jung, S., Kim, H., Jeoung, J., & Hong, T. (2024). Reinforcement learning-based optimal scheduling model of battery energy storage system at the building level. Renewable and Sustainable Energy Reviews, 190, Article 114054. https://doi.org/10.1016/j.rser.2023.114054

[38] Shang, Y., Wu, W., Guo, J., Ma, Z., Sheng, W., Lv, Z., & Fang, H. (2020). Stochastic dispatch of energy storage in microgrids: An augmented reinforcement learning approach. *Applied Energy, 261*, Article 114423. https://doi.org/10.1016/j.apenergy.2019.114423

[39] Kang, H., Jung, S., Kim, H., Jeoung, J., & Hong, T. (2024). Reinforcement learning-based optimal scheduling model of battery energy storage system at the building level. *Renewable and Sustainable Energy Reviews, 190*, Article 114054. https://doi.org/10.1016/j.rser.2023.114054

[40] Wu, J., Li, S., Yu, Y., Guan, Y., Wei, B., & Chen, Z. (2026). Deep reinforcement learning-based multi-objective energy management system for microgrids under flexible energy market. *Applied Energy, 406*, Article 127223. https://doi.org/10.1016/j.apenergy.2025.127223

[41] Haarnoja, T., Zhou, A., Hartikainen, K., Tucker, G., Ha, S., Tan, J., Kumar, V., Zhu, H., Gupta, A., Abbeel, P., & Levine, S. (2019). Soft actor-critic algorithms and applications. *The International Journal of Robotics Research, 38*(2–3), 361–378. https://doi.org/10.1177/0278364919899452

[42] Fan, J., & Wang, H. (2023). *Deep reinforcement learning for community battery scheduling under uncertainties of load, PV generation, and energy prices*. arXiv. https://arxiv.org/abs/2312.03008

[43] Li, Y., Wu, J., & Pan, Y. (2024). Deep reinforcement learning for online scheduling of photovoltaic systems with battery energy storage systems. *Intelligent and Converged Networks, 5*(1), 28–41. https://doi.org/10.23919/ICN.2024.0003

[44] Kang, H., Jung, S., Kim, H., Jeoung, J., & Hong, T. (2024). Reinforcement learning-based optimal scheduling model of battery energy storage system at the building level. Renewable and Sustainable Energy Reviews, 190, Article 114054. https://doi.org/10.1016/j.rser.2023.114054

[45] Hu, C., Cai, Z., Zhang, Y., Yan, R., Cai, Y., & Cen, B. (2022). A soft actor-critic deep reinforcement learning method for multi-timescale coordinated operation of microgrids. *Protection and Control of Modern Power Systems, 7*, Article 29. https://doi.org/10.1186/s41601-022-00252-z

[46] Selim, A., Mo, H., Pota, H., & Dong, D. (2026). Adaptive optimization of BESS and grid set points: A model-free framework for energy management under dynamic tariff pricing. *Energy, 346*, Article 140148. https://doi.org/10.1016/j.energy.2026.140148

[47] Zhang, Y., Gatsis, N., & Giannakis, G. B. (2013). Robust energy management for microgrids with high-penetration renewables. *IEEE Transactions on Sustainable Energy, 4*(4), 944–953. https://doi.org/10.1109/TSTE.2013.2255135

[48] Khodaei, A., Bahramirad, S., & Shahidehpour, M. (2015). Microgrid planning under uncertainty. *IEEE Transactions on Power Systems, 30*(5), 2417–2425. https://doi.org/10.1109/TPWRS.2014.2361094

[49] Zhang, B., Li, Q., Wang, L., & Feng, W. (2018). Robust optimization for energy

transactions in multi-microgrids under uncertainty. *Applied Energy, 217*, 346–360. https://doi.org/10.1016/j.apenergy.2018.02.121

[50]Valibeygi, A., Habib, A. H., & de Callafon, R. A. (2019). Robust power scheduling for microgrids with uncertainty in renewable energy generation. In *2019 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)* (pp. 1–5). IEEE. https://doi.org/10.1109/ISGT.2019.8791649

[51]Zhong, J., Li, Y., Cao, Y., Tan, Y., Peng, Y., Zhou, Y., Nakanishi, Y., & Li, Z. (2023). Robust coordinated optimization with adaptive uncertainty set for a multi-energy microgrid. *IEEE Transactions on Sustainable Energy, 14*(1), 111–124. https://doi.org/10.1109/TSTE.2022.3203299

[52]Kou, P., Liang, D., & Gao, L. (2018). Stochastic energy scheduling in microgrids considering the uncertainties in both supply and demand. *IEEE Systems Journal, 12*(3), 2589–2600. https://doi.org/10.1109/JSYST.2016.2580609

[53]Wiesemann, W., Kuhn, D., & Sim, M. (2014). Distributionally robust convex optimization. *Operations Research, 62*(6), 1358–1376. https://doi.org/10.1287/opre.2014.1314

[54]Rahimian, H., & Mehrotra, S. (2022). Frameworks and results in distributionally robust optimization. *Open Journal of Mathematical Optimization, 3*, Article 4. https://doi.org/10.5802/ojmo.15

[55]Nilim, A., & El Ghaoui, L. (2005). Robust control of Markov decision processes with uncertain transition matrices. *Operations Research, 53*(5), 780–798. https://doi.org/10.1287/opre.1050.0216

[56]Derman, E., Geist, M., & Mannor, S. (2021). Twice regularized MDPs and the equivalence between robustness and regularization. *Journal of Machine Learning Research, 22*(74), 1–39.

[57]Antoniadou-Plytaria, K., Steen, D., Le, A. T., Carlson, O., & Fotouhi Ghazvini, M. A. (2021). Market-based energy management model of a building microgrid considering battery degradation. *IEEE Transactions on Smart Grid, 12*(2), 1794–1804. https://doi.org/10.1109/TSG.2020.3037120

[58] Xu, B., Zhao, J., Zheng, T., Litvinov, E., & Kirschen, D. S. (2018). Factoring the cycle aging cost of batteries participating in electricity markets. *IEEE Transactions on Power Systems, 33*(2), 2248–2259. https://doi.org/10.1109/TPWRS.2017.2733339

[59] García-Miguel, P. L., García-Torres, F., & Bordons, C. (2022). A review on the degradation implementation for the operation of battery energy storage systems. *Batteries, 8*(9), Article 110. https://doi.org/10.3390/batteries8090110

[60] Zhao, Y.-B., Dong, X.-J., Shen, J.-N., & He, Y.-J. (2024). Simultaneous sizing and scheduling optimization for PV-wind-battery hybrid systems with a modified battery lifetime model: A high-resolution analysis in China. *Applied Energy, 360*, Article 122812. https://doi.org/10.1016/j.apenergy.2024.122812

[61] Lundmark, L., Golzar, F., Guedez, R., & Astaneh, M. (2026). A validated forecasting model for optimizing dispatch schedules of PV-battery system in grid-connected apartment buildings. *Applied Energy, 404*, Article 127123. https://doi.org/10.1016/j.apenergy.2025.127123

[62] Ju, C., Wang, P., Goel, L., & Xu, Y. (2018). A two-layer energy management system for microgrids with hybrid energy storage considering degradation costs. *IEEE Transactions on Smart Grid, 9*(6), 6047–6057. https://doi.org/10.1109/TSG.2017.2703126

[63] Shi, Y., Xu, B., Tan, Y., & Zhang, B. (2018). A convex cycle-based degradation model for battery energy storage planning and operation. In *2018 Annual American Control Conference (ACC)* (pp. 4590–4596). IEEE. https://doi.org/10.23919/ACC.2018.8431814

[64] Craparo, E. M., Karatas, M., & Singham, D. I. (2017). A robust optimization approach to hybrid microgrid operation using ensemble weather forecasts. *Applied Energy, 201*, 135–147.

[65] Valibeygi, A., Habib, A. H., & de Callafon, R. A. (2019). Robust power scheduling for microgrids with uncertainty in renewable energy generation. *arXiv*. https://arxiv.org/abs/1902.07927

[66] Jiang, J., Zhou, R., Xu, H., Wang, H., Wu, P., Wang, Z., & Li, J. (2022). Optimal sizing, operation strategy and case study of a grid-connected solid oxide fuel cell microgrid. *Applied Energy*, *307*. https://doi.org/10.1016/j.apenergy.2021.118214

[67] Zhang, Y., Ma, T., Elia Campana, P., Yamaguchi, Y., & Dai, Y. (2020). A techno-economic sizing method for grid-connected household photovoltaic battery systems. *Applied Energy*, *269*. https://doi.org/10.1016/j.apenergy.2020.115106

[68] Greenpeace. (2023). RE10X10 Corporate Green Energy Initiative 2023 Annual Report. Retrieved from https://cloud.greentw.greenpeace.org/campaign-climatere10x10

[69] Minh, N. Q., Linh, N. D., & Khiem, N. T. (2024). A mixed-integer linear programming model for microgrid optimal scheduling considering BESS degradation and RES uncertainty. *Journal of Energy Storage*, *104*. https://doi.org/10.1016/j.est.2024.114663

[70] Hu, Z., & Hong, L. J. (2013). *Kullback-Leibler divergence constrained distributionally robust optimization*. Optimization Online.

[71] Van Hoof, H., Chen, N., Karl, M., Van Der Smagt, P., & Peters, J. (2016). Stable reinforcement learning with autoencoders for tactile and visual data. In *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 3928–3934). IEEE.

[72] Fujimoto, S., Meger, D., & Precup, D. (2019). Off-policy deep reinforcement learning without exploration. In *International Conference on Machine Learning* (pp. 2052–2062). PMLR.

[73] Rockafellar, R. T., & Wets, R. J.-B. (2009). *Variational analysis* (Vol. 317). Springer Science & Business Media.

[74] Fujimoto, S., van Hoof, H., & Meger, D. (2018). Addressing function approximation error in actor-critic methods. In J. Dy & A. Krause (Eds.), Proceedings of the 35th International Conference on Machine Learning (Vol. 80, pp. 1587–1596). PMLR. https://proceedings.mlr.press/v80/fujimoto18a.html

[75] Dugas, C., Bengio, Y., Bélisle, F., Nadeau, C., & Garcia, R. (2000). Incorporating second-order functional knowledge for better option pricing. In T. K. Leen, T. G. Dietterich, & V. Tresp (Eds.), *Advances in Neural Information Processing Systems 13* (pp. 472–478). MIT Press.

[76] Vinothine, S., Widanagama Arachchige, L. N., Rajapakse, A. D., & Kaluthanthrige, R. (2022). Microgrid Energy Management and Methods for Managing Forecast Uncertainties. In *Energies* (Vol. 15, Issue 22). MDPI. https://doi.org/10.3390/en15228525

[77] Sorour, A., Fazeli, M., Monfared, M., Fahmy, A. A., Searle, J. R., & Lewis, R. P. (2022). MILP Optimized Management of Domestic PV-Battery Using Two Days-Ahead Forecasts. *IEEE Access*, *10*, 29357–29366. https://doi.org/10.1109/ACCESS.2022.3158303

[78] Gomes, I. L. R., Ruano, M. G., & Ruano, A. E. (2023). MILP-based model predictive control for home energy management systems: A real case study in Algarve, Portugal. *Energy and Buildings*, *281*, 112774. https://doi.org/10.1016/j.enbuild.2023.112774

[79] Sharma, V., Cortes, A., & Cali, U. (2021). Use of forecasting in energy storage applications: A review. *IEEE Access*, *9*, 114690–114704. https://doi.org/10.1109/ACCESS.2021.3103844

[80] Zonjee, T., & Torbaghan, S. S. (2023). Energy Storage Arbitrage in Day-Ahead Electricity Market Using Deep Reinforcement Learning. *2023 IEEE Belgrade PowerTech, PowerTech 2023*. https://doi.org/10.1109/PowerTech55446.2023.10202674

[81] Ali, K. H., Sigalo, M., Das, S., Anderlini, E., Tahir, A. A., & Abusara, M. (2021). Reinforcement learning for energy-storage systems in grid-connected microgrids: An investigation of online vs. offline implementation. *Energies*, *14*(18), 5688. https://doi.org/10.3390/en14185688

[82] Gurobi Optimization, LLC. (2023). *Gurobi optimizer reference manual*. https://www.gurobi.com